

A Students who has just entered the portals of colleges finds it difficult to understand the research taught to him/her this difficulty is mainly due to understand standard references. While preparing this book "PHYSIO-CHEMICAL ANALYSIS IN SELECTED SITES AND MOLECULAR STUDIES WITH CONSERVATION STATUS OF FRESH WATER STREAM FISHES IN PALANI HILLS OF SOUTHERN INDIA "has been prepared to serve the special purpose of research students. This book explains the basic research methodology and Techniques in a simplified manner. This book can be used for PG and research scholar of Zoology students, who are looking for a quick revision before their research.



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"THE HAND BOOK OF MINI PROJECT

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PREFACE

A Students who has just entered the portals of colleges finds it difficult to understand the research taught to him/her this difficulty is mainly due to understand standard references. While preparing this book **“PHYSIO-CHEMICAL ANALYSIS IN SELECTED SITES AND MOLECULAR STUDIES WITH CONSERVATION STATUS OF FRESH WATER STREAM FISHES IN PALANI HILLS OF SOUTHERN INDIA”** has been prepared to serve the special purpose of research students. This book explains the basic research methodology and Techniques in a simplified manner. This book can be used for PG and research scholar of Zoology students, who are looking for a quick revision before their research.

This book support to the research of Zoology students to get confidence and also reference just before attending Dissertation Examination of PG AND MPhil scholar and other Ph.d students. This Book has been written to research students and preparatory the field of life sciences studies, No such book which cover suitable and needed explanation of research techniques and Practice. This book is useful for the preparation of their dissertation and thesis of state Universities and College students integrated courses field of research.

Dr. S. KALAIMANI

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And last but not least I wish to extend my sincere thanks to **All My Friends and Relatives** for their co-operation and support in this Mini project work to make success.

Mr. S.KALAIMANI

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I.INTRODUCTION

Western Ghats of South India is a renowned UNESCO world Heritage site and is one of the ‘Hotspots’ of biological diversity, popularly known as “Great Escarpment of India “Western Ghats of South India is well known for the rich freshwater fish fauna with a high level of endemism. However a major part of the fauna diversity of Western Ghats is threatened by human intervention and invasive fish species. The knowledge of the diversity and distribution of the fish fauna is essential for designing and implementing conservation strategies. In the present day of study small palani hills streams located within the latitudinal range of 10.10N – 10.44 N, longitudinal range of 77.3 E to 78.68 E within the altitudinal range of 250m – 2050m mean sea level (MSL) that flows Southeast and South West direction covering about 90.55 sq km area from origin were studied in Palani hills.

Small Palani hills streams generally have rough, rocky, boulder and pebbled substratum which affect the occurrence, distribution and productivity of fish fauna. All the fresh water resources of the Palani hills region. Compounds a very rich and colorful ichthyofauna, different species of fishes have been abundantly habited in different streams in the study area. Fish distribution and abundance from different parts of Western Ghats have been evaluated by different authors. Detailed account of freshwater fishes has been provided by Day. Further, investigations on the freshwater fish fauna of Western Ghats were initiated by Hora and Hora and Law. Silas listed 25 fish species from the Anamalai Hills and 10 species from the Nelliampathi Hills. Arunachalam and Manimekalan reported economically important and cultivable fishes of the Nilgiri biosphere reserve later he described the assemblage structure of stream fishes in the Western Ghats. Meanwhile, Manimekalan and Das described a new species, *Glyptothorax davissinghi* (Pisces: Sisoridae), a new catfish from Nilambur, Nilgiri Biosphere. Manimekalan and Singh recorded *Schismatorhynchus* (Nukta) (Nukta (Sykes) (Pisces: Cyprinidae) from Moyar River. Later, Arunachalam et al., reported the occurrence of *Neolissochilus wynaadensis*, from Karnataka. Earlier, Biju et al., recorded *Puntius filamentosus* and *Puntius melanampyx* (Day) in Orukomban and Thelikal respectively. Manimekalan and Arunachalam

(2002) rediscovered the critically endangered air – breathing catfish *Clarias dayi* Hora (Pisces: Claridae) in Mudumalai Wildlife Sanctuary. Johnson and Arunachalam reported the diversity, distribution and assemblage structure of fishes in streams of southern Western Ghats. Knight et al., described a new species of barb *Puntius nigripinnis* (Teleostei: Cyprinidae (from southern Western Ghats. Rema Devi and Menon described *Horababiosa palaniensis*, Cyprinid fish from Palani Hills, Western Ghats. Recently, Arunkumar et al. Reported nearly 37 species in the Cauvery River System. Meanwhile, Mogalekar of Tamil Nadu with the growing international trade of ornamental fishes, some of the fishes like *Puntius denisonii*, *Tetraodon travancoricus*, *Horabagrus nigricollaris* and several species of the genus *Puntius*, *Danio*, *Garra*, *Loaches*, *Bagrids* and *Cichlids* are in great demand in domestic as well as international market as aquarium fish. These fishes are being collected from natural habitats and exported resulting into drastic in their natural population. Several studies have indicated that breeding of *Nemacheilus* species *Puntius denisonii* is warranted as over exploitation of wild stock of these fishes could be the main reason for their diminution in the wild. However, the primary constraint with the endemic ornamental fish species is the loss of natural bright colour under captivity. Therefore, few of the endemic ornamental species have been identified for captive breeding and aquaculture *Puntius denisonii* and *Danio malabaricus*, along with other species like *P. Fasciatus*, *P. melanostigma*, *P. filamentosus*, *P. arulius*, *Tetraodon travancoricus*, *Chela dadyburjori*, *Horabagrus brachysoma*, *H.nigricollaris* and *Nemacheilus* as over-exploitation of wild stock of these highly-priced fishes can lead to their extinction.

It has been now well established that fish diversity is determined by the geographical location, ecological condition and physiochemical nature of the aquatic habitat. Freshwater fishes, for example, may be the most threatened group of vertebrates on earth after amphibians and the global extinction rate of fishes is believed to be in excess of higher vertebrates (Bruton, 1995) However, conservatory measures to mitigate the impact of the pressures have not only been slow but also inadequate and as a result many of the aquatic species are declining rapidly. The main causes behind the loss of biodiversity in freshwater are habitat degradation and fragmentation, exotic species introduction, water

diversions, pollution and global climate impacts (Gibbs, 2000). The Indian subcontinent, occupying a position at the confluence of three biogeography realms, viz., the Palaearctic, Afro-Tropical and Indo-Malayan, exhibits a great variety of ecological habitats, harboring rich ichthyofaunal diversity, comprising about 2500 species of which 930 species are freshwater inhabitants and 1570 are marine. The Indian species represent about 8.9% of the known fish species of the world. Jayaram (2010) listed 852 freshwater species of fishes under 272 genera, 71 families and 16 orders, including both primary and secondary freshwater fishes from India, Bangladesh, Myanmar, Nepal, Pakistan and Sri Lanka. Freshwater fishes are a poorly studied group since information regarding distribution, population dynamics and threats is incomplete, and most of the information available is from a few well-studied locations only (Sabuj Kumar chaudhuri 2010).

Conservation and management of biodiversity relies on effectively monitoring rare (or) patchily distributes species across large area (Bogich et al., 2008). The presence of species – specific DNA fragments without isolating target organisms (or) their parts from the sample (Ficetola et al., 2008). Determines how well c DNA can serve as a proxy for directly observing organisms is an area of active research that will influence Molecular characterization research method should be applied to biological conservation (Foot et al., 2012) High concentration DNA such as PCR product and fresh tissue produce the greatest risk of contamination that cannot be objectively distinguished from real DNA detection (champlot et al., 2010) C DNA can persistent for thousands of years depends on starting concentration and degradation conditions (Levy – Booth et al., 2007). These feature of Coi detection create uncertainly whose characterization and appropriate use required better understanding of gene in four domain: Origin, state and fate. The origin of gene describe its physiological source, commonly hypothesized to be feces, Urine, gameter, skin and decomposition (Caldwell et al., 2011) the state of gene describe its mutable physical forms such as particle band (or) freely dissolved DNA molecules (Turnover et al., 2014). The fate of gene describes its extracellular DNA fragments too small for identification (Barnes et al., 2014). In the present study we aimed to gain new understanding about the transport of C

DNA shed by fish, specifically its relative concentration in 15 alternative locations.

The growing international trade of ornamental fishes, some of the fishes like *Puntius denisonii*, *Tetraodon travancoricus*, *Horabagrus nigricollaris* and several species of the genus *puntium*, *Danio*, *Garra*, *Loaches*, *Bagrid* and *Cichids* are in great demand in domestic as well as international market as aquarium fish, these fishes are being collected from natural habitat and exported result into drastic decline in their natural population. Several studies have indicated that breeding of *Nemacheilus* species *Puntius denisonii* is warranted over exploitation of wild stock of these fishes could be the main reason for their diminution in the wild.

However, the primary constraint with the endemic ornamental fish species is the loss of natural bright colour under captivity. Therefore few of the endemic ornamental species have been identified for captive breeding and agriculture *Puntius denisonii* and *Danio malabaricus* along with other species like *P. fasciatus*, *P. melano stigma*, *P. filamentosus*, *P. arulius*, *Tetraodon travancoricus*, *Chela dadyburjori*, *Horabagrus brachysoma*, *H. nigricollaris* and *nemacheilus* as over exploitation of wild stock of these highly – priced fish can lead to their extinction.

Molecular techniques offer much more reliable results for resolving problems in systematic and the classification of taxa. Technological advances in molecular biology and biochemistry have also led to the development of a variety of genetic markers that can be used to address questions of relevance to the management and conservation of fish species (Ferguson & Danzmann 1998). The necessity to detect genetic variation has fueled the development of novel marker systems in fisheries biology. Influential and advanced methods such as microsatellite analysis and DNA sequencing prove to be among the best in the detection of variation but are time consuming and expensive. In this study, our goal is to provide a simple, fast and non-expensive molecular method to discriminate among species of freshwater fishes by employing DNA barcoding technique. Thus, this study employs the analysis of the cytochrome b (Cyt b) mitochondrial DNA (mtDNA) gene segment in an effort to find genetic markers

that are diagnostic among freshwater fish species, in other words markers that could distinguish or revealed the unique genetic identity of a particular species. It has several advantages over nuclear genome for diagnostic studies of animals because of greater abundance in sample extracts and higher copy number making the amplification of mtDNA segment to be a relatively sensitive procedure (Sumathi et al., 2015).

Several mitochondrial genes have been used repeatedly to resolve phylogenies of cypriniform fishes. Cytochrome b is the workhorse of molecular phylogenetics in vertebrates and as expected is the most widely used gene for cypriniform systematics. Cytochrome b (Cyt b) is one of the most important protein encoding genes on the heavy strand of the mtDNA molecule. It is one of the best known proteins that make up complex III of the mitochondrial phosphorylation system and is the only one encoded by the mitochondrial genome. The Cyt b gene is used as an important utility tool in studies of legal medicine and molecular evolution (Prusak et al., 2004; Abou-Hadeed et al., 2011). This gene has been completely sequenced or partially sequenced for many species of mammals, birds (Bravi et al., 2004; Andrzej and Kamila, 2005), reptiles, amphibian and fishes (Chow et al., 1993; Ram et al., 1996; Quinterio et al., 1998; Lidstrom, 1999; Parson et al., 2000) and also some invertebrates (Lee et al., 2009a). It has been widely used in molecular marker techniques for many relevances to assess intra- or interspecies genetic diversity, genetic variation, phylogeography, species and hybrid identification, phylogeny in numerous species and genera, population genetic structure, conservation, and demographic history (Hsu et al., 2009; Li et al., 2009; Ma et al., 2010; Thangaraj and Lipton, 2010).

Our previous study provided information of the physical–chemical and morphological properties, including morphometrics and meristic variations that have been reported in Cyprinidae species, suggesting significant heterogeneity among selected species. However, the variation of morphological characteristics is probably influenced by genetic and environmental factors, so discrimination of populations based on morphological variation should be confirmed by evidence to verify that morphological variation reflects a true degree of reproductive isolation, rather than environmental isolation (Tzeng, 2004; Tzeng

et al., 2008). Therefore, in the present study, we used mtDNA Cyt b to investigate the genetic discrimination of the six species of Cyprinidae family in Southern Western Ghats of Peninsular India and to provide a preliminary assessment of genetic diversity patterns of Cyprinidae fishes in order to improve knowledge. Our findings have potential implications for the management, conservation, and stock identification as important resources for the fisheries industry. Such information for this family will be useful in the management of different species of Cyprinidae for Good Agriculture Practices (GAP) in farming technology, as well as providing the basis for Good Manufacturing Practices (GMP).

The Western Ghats mountain range extends along the west coast of India and is crisscrossed with streams, which form the headwaters of several major rivers providing water to the plains of peninsular India. The Ghats represent a globally critical ecosystem and biodiversity hotspot (Myers et al. 2000). Freshwater fish diversity is very high, with around 288 species and a high rate of endemism (>50%) (Daniels 2002; Dahanukar 2004). The southern Western Ghats that comprise the Ashambu (Agasthyamala) Hills landscape in Kerala form a biogeographically unique ecoregion (Nair 1991). As with several other taxa, this region is very rich in fish species diversity and endemism (Bossuyt et al. 2004). West flowing rivers south of the Shenkottah/Ariyankavuppass have not been extensively surveyed for fish diversity. Most surveys to date have been conducted either in easily accessible sites or in limited localities along individual rivers. In this paper we provide a systematic, updated checklist of freshwater fishes across five important rivers namely, Kallada, Ithikkara, Vamanapuram, Karamana and Neyyar, based on the available literature, secondary data and field-sampling along these rivers in the Ashambu Hills landscape. Habitat preferences and anthropogenic threats have been listed for each species included in the checklist, along with ecological and threat information. This can facilitate the identification of conservation priorities for freshwater fish diversity in the region.

Aeromonas species are gram negative, non-sporulating, aerobic and facultative anaerobic bacteria, which are widely distributed throughout the world. They are classified under taxa Bacteria; Proteobacteria;

Gammaproteobacteria; Aeromonadales; family-Aeromonadaceae; genus-Aeromonas. They are oxidase and catalase positive, reduce nitrate to nitrite, and ferment D-glucose. Aeromonads are related to disease problems in humans and are also pathogenic for aquatic and terrestrial animals (Austin and Adams, 1996). Aeromonas species are responsible for gastroenteritis, chronic diarrhoea, wound infections, respiratory tract infections, peritonitis, urinary tract infections, and septicemia. In human *A. hydrophila*, *A. caviae*, and *A. veronii* are the predominating species responsible for diseases, whereas *A. eucrenophila*, *A. popoffii* and *A. culicicola* have never been found in clinical samples. Some Aeromonas species are associated with a wide variety of diseases in cold- and warmblooded animals including fish, frogs, water buffaloes, reptiles, birds and cattle. *A. veronii*, *A. hydrophila*, and *A. salmonicida* are capable of causing septicemia in freshwater and marine fish. Aeromonas species have been responsible for haemorrhagic septicemia or motile Aeromonas septicemia, infectious dropsy, red mouth disease, red pest disease in fish and reported to be associated with Epizootic Ulcerative Syndrome (EUS) in South East Asian Countries and India (Roberts et al., 1992). Clinical and environmental Aeromonas species isolates secrete many extracellular products, such as hemolysins, enterotoxins, and proteases. Some of the genes in Aeromonas contribute to its toxicity. The cytotoxic enterotoxin gene (*act*), a single-chain polypeptide, is one of the primary genes that make this species pathogenic. Heat-labile cytotoxic enterotoxins (*Alt*) and heat-stable cytotoxic enterotoxins (*Ast*) also contribute to toxicity (Albert et al., 2000).

Studies conducted by Kuhn et al., (1997) showed that some isolates of a given species produce virulence factors more frequently than others. These findings indicate that the virulence within the genus Aeromonas might be a clonal property and only some clones may be responsible for progressive disease. Since Aeromonas hydrophila was first recognized as an important pathogen for aquatic animals and significant opportunistic pathogen for humans, thus having public health significance, many efforts were dedicated to find methods for accurate identification, classification and typing of species belonging to this genus.

Aeromonads species are known to be phenotypically, serologically and genetically quite diverse and the conventional methods of identifying these microorganisms give contradictory results. Biochemical tests are known to be time consuming and sometimes lead to confusion in diagnosis of species of pathogens. Again serological tests are sometimes handicapped by lack of specificity of antisera and cross-reactivity. It appears that a rapid and accurate identification of cenospecies of Aeromonads would be useful diagnostic tools in clinical and veterinary laboratories. Attempts have been made by application of molecular techniques for prompt identification, classification and characterization of microbial pathogens. Although different molecular methods like Polymerase Chain Reaction (PCR), Randomly Amplified Polymorphic DNA (RAPD), Restriction Fragment Length Polymorphism (RFLP), and Ribo typing etc. have been applied to identify and characterize the pathogens, none have been found suitable for accurate diagnosis of different *Aeromonas* species/ strains/ isolates.

Again, there is a need to develop and validate molecular genotyping techniques so as to create a genomic fingerprint database, which can be helpful to develop a surveillance system to monitor the genetic variability and molecular epidemiology of *Aeromonas* pathogens, which has been a major area of concern. This genotyping information would be useful to identify prevalent strains causing disease in specific groups/ regions, to identify new virulent strains introduced into the population and to track potential sources of these virulent strains. In molecular epidemiological analysis, gene sequencing has been largely used as the gold standard to study strain variation for many viruses. Bacterial genomes being much larger and more complex than viral genomes, sequencing of a single gene or a fragment of a gene from a bacterial genome is unlikely to represent the variability present in the whole genome. That is why methods that can characterize large portions of the genome are preferable to study the molecular epidemiology of bacterial genome. A PCR-based method called "Repetitive sequence- PCR (Rep-PCR) developed in the early 90's, has been used, which allows typing of bacterial organisms in a single PCR reaction. MLST (Multi Locus Sequence Typing) amplifies number genetic loci of any genome and compares their sequences.

West Bengal is the only state of India that extends from the Himalaya to the Bay of Bengal and located between Latitude 21°38' N to 27°10' N and Longitude 85°38' E to 89°50' E. A large portion of the state occupies the transitional zones between the Himalayas in the north and the Chhotanagpur plateau in the west to the plains of the Ganga-Brahmaputra delta in the southern and eastern sections. West Bengal is enriched with 6.08 lakh hector of freshwater fisheries resources in the form of Ponds and Tanks (2.88 lakh hector), Beels and Boar (0.41 lakh hector), Reservoirs (0.27 lakh hector), 22 river drainage basins (1.72 lakh hector) and Canals (0.80 lakh hector). Basins of river Ganga (81% of area), river Brahmaputra (12%), river Subarnarekha (4%) and two small coastal rivers (3%) provide an important domain for freshwater fish biodiversity in West Bengal. India is an exceptional hotspot of freshwater fish diversity with a high degree of endemism contributing to the World's biological resources. Understanding of species and communities reveal crucial facts necessary to the management of ecosystem and habitat. Main causes of the decline of native fish species are habitat loss and species introductions. The conservation of the freshwater fishes has never been adequately addressed in India which has been mainly due to lack of scientific data and perception about the criteria to be used for the characterization and designating conservation status of threatened fishes. Knowledge of the species composition and distribution patterns of fishes are fundamental for conservation and management of native fish fauna. In this concern an attempt has been made to acquire information on fundamental variables like taxonom, distribution, environment, and maximum size, pattern of human use and conservation status of freshwater fishes of West Bengal to standardize the clean taxonomic nomenclature to agreed-upon standards.

Genetic diversity, the level of biodiversity, refers to the total number of genetic characteristics in the genetic makeup of a species. Genetic diversity (Synonymous to genetic variability, intraspecific diversity and biological diversity on gene level) represents the existence of variants or alleles of individual genes resulting from alteration of DNA sequences. Genetic diversity serves as a way for different subpopulations of a species to adapt the changing environments. With more variation, it is more likely that some individuals in a

population will possess variations of alleles that are suited for the changing environment and these individuals are more likely to survive to produce offspring bearing that allele. The population will continue for more generations because of the success of these individuals. Genetic diversity between individuals in a population is designated as polymorphism and the variation within the individuals as heterozygosis.

The development of molecular genetics technique has made possible to decipher the genetic constituent of fish that has been revolutionized our ability to understand the genetic divergence at subspecies level. Nowadays molecular markers are used in order to discriminate genetic diversity. The randomly amplified polymorphic DNA (RAPD) analysis is a technique based on the polymerase chain reaction. (PCR), used to detect DNA polymorphism, by amplifying discrete regions of genome with short oligonucleotide primers of arbitrary sequence (Welsh and McClelland, 1990; Williams et al., 1990). This technique was used extensively to detect genetic diversity in animals (Cushwa and Medrano, 1996). RAPD technique is considered to be one of the best techniques for taxonomic systematic study of various organisms (Garg et al., 2009a, 2009b). RAPD markers have many applications like gene mapping, population genetics, molecular evolutionary genetics and plant and animal breeding (Bardacki and Skibinski, 1994; Ertas and Seker, 2005). Intra and inter population variation is essential for better survival of individuals against the changing environment (Rayman et al., 1995).

RAPD was used to evaluate genetic diversity in various groups of fishes like guppy (Dinesh et al., 1996), catfish (Liu et al., 1998; Almeida et al., 2003) common carp (Bartfi et al., 2003) and Indian major carps (Barman et al., 2003). Beside this the technique was also used to study the phylogeny of freshwater fishes such as sturgeon by Comincini et al., (1998) and Angullidae by Lehmann et al., (2000). RAPD technique was found to be more sensitive than mitochondrial DNA analysis which failed to detect DNA polymorphism in the populations of tilapia (Caapilai, 1990; Seyoum and Komfield, 1992) while Bardacki and Skibinski (1994) reported intraspecific variation in tilapia using RAPD technique. Several authors used RAPD as a tool for the assessment of genetic variability. Apostolodis et al., (2011) reported the genetic diversity in

the Balkan trout population (*Salmo trutta*). RAPD tool was also used in molecular typing of bacterial strains which affect the fish population. In the same way Castro et al. (2012) also reported genetic variability of bacterial strain, *Edwardsiella tarda* sampled from the fish namely Senegalese sole, *Solea senegalensis* (Kaup) from the Spain. The study suggested that RAPD markers can be used for stock selection for selective breeding programmed. However to keeping in view the above mentioned facts the RAPD technique was used in the present study to evaluate the genetic diversity in *Colisa lalia*. The human interferences on aquatic habitats has added a sense of urgency to investigate the possible link between habitat destruction and the breakup of gene pools, redistribution of genetic variability and loss of sustainability. Conservation of genetic diversity of fishes is fundamentally important in protection of aquatic biodiversity, and preservation of aquatic diversity is imperative for sustainable management of fisheries management. The loss of populations may be more fatal than loss of species (Mayer, 1997). Since, there is no information available regarding the genetic variability of *C. Lalia*, the current study was planned to investigate the genetic variations in *C. lalia* RAPD as a molecular marker.

II.OBJECTIVE OF THE STUDY

- 1.To inventories the diversity, seasonality and distributional pattern of freshwater fishes in palani hills
- 2.To assess the conservation status of freshwater fishes in palani hills

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III. REVIEW OF LITERATURE

Allendorf et al., 1987; Wimberger, 1992, In general, fish demonstrate greater variances in morphological traits both within and between populations than any other vertebrates, and are more susceptible to environmentally induced morphological variations. **Murta, 2000; Costa et al., 2003**, The phenotypic discreteness suggests a direct relationship between the extent of phenotypic divergence and geographic separation, which indicated that geographic separation, is a limiting factor to migration among stocks. Both morphometric and meristic characters are important in morphological studies. Here in the current study all 22 morphometric characters were significantly different in the populations. Studies already showed that morphometric characters are more suitable than meristic characters to describe the intra specific variation. **Hinder and Johnson, 1993; Peres-Neto and Magnan, 2004**, meristic characters were considered to be fixed in early embryonic life of the individual and remain unchanged thereafter. Thus, they respond to environmental factors only a short period of time during embryonic development. In contrast, morphometric characters varied according to the changing environmental conditions of the habitat throughout their life and the phenotypic plasticity were shown in many freshwater fish species.

Pakkasamaa et al., (1998), also reported the pronounced interspecific variation as compared to intraspecific variations in four land locked salmonid species. **Cakmak and Alp (2010)**, reported morphological differences among the Mesopotamian spiny eel *Mastacembelus mastacemelus* populations. The significant divergence in morphological and meristic traits revealed that intraspecific variation existed among the specimens of *C. lalia*. Therefore, multiracial composition of *C. lalia* across the environmental gradient cannot be ruled out. **Welsh and McClelland, 1990; Williams et al., 1990**, The development of molecular genetic technique has made possible to decipher the genetic constituent of fish that has been revolutionized our ability to understand the genetic divergence at subspecies level. Nowadays molecular markers are used in order to discriminate genetic diversity. The randomly amplified polymorphic DNA (RAPD) analysis is a technique based on the polymerase chain reaction (PCR), used to detect DNA polymorphism, by amplifying

discrete regions of genome with short oligonucleotide primers of arbitrary sequence. **Garg et al., 2009a, 2009b**, This technique was used extensively to detect genetic diversity in animals (Cushwa and Medrano, 1996). RAPD technique is considered to be one of the best techniques for taxonomic systematic study of various organisms **Bardakci and Skibinski, 1994; Ertas and Seker, 2005**, RAPD markers have many applications like gene mapping, population genetics, molecular evolutionary genetics and plant and animal breeding. **Rayman et al., 1995**, Intra and inter population variation is essential for better survival of individuals against the changing environment. **Barman et al., 2003**, RAPD was used to evaluate genetic diversity in various groups of fishes like guppy (Dinesh et al., 1996, catfish (Liu et al., 1998; Almedia et al., 2003) common carp (Bartfiet et al., 2003) and Indian major carps. **Anguillidae by Lehmann et al., 2000**, beside this the technique was also used to study the phylogeny of freshwater fishes such as sturgeon. **Capilai, 1990; Seyoum and Komfield, 1992**, RAPD techniques was found to be more sensitive than mitochondrial DNA analysis which failed to detect DNA polymorphism in the populations of tilapia while Bardakci and Skibinski (1994) reported intraspecific variation in tilapia using RAPD technique. Several authors used RAPD as a tool for the assessment of genetic variability. **Apostolodis et al., (2011)**, reported the genetic diversity in the Balkan trout population (*Salmo trutta*). RAPD tool was also used in molecular typing of bacterial strains which affect the fish population. **Castro et al., (2012)**, In the same way also reported genetic variability of bacterial strain, *Edwardsiella ictaluri* sampled from the fish namely Senegalese sole, *Solea senegalensis* (Kaup) from the Spain. The study suggested that RAPD markers can be used for estimation of genetic variation and these genetic variation data can be used for estimation of genetic variation and these genetic variation data can be used for stock selection for selective breeding programmed. **Welsh et al., 1991**, In the present study, the analysis of RAPD data screened genetic variations among the populations of *C. lalia*. The number of amplified bands ranged between 282-420 with size ranging from 150 bp-2500 bp. Actually, the number and size of band generated strictly depends upon primer sequence and the DNA template.

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Ward, Hanner and Hebert (2009), Hubert et al., (2008) found average K2P values of 0.27% (conspecific) and 8.37% (congeneric), discriminating 93% of the species while studying freshwater fishes from Canada. In another study 93% of freshwater fishes were discriminated from Mexico and Guatemala showing average K2P values of 0.45% in conspecific and 5.1% in congeneric (Valdez-Moreno, Ivanova, Elias-Guitierrez. **Hubert et al., (2007)**, similarly pointed to a recent radiation of some groups of freshwater fishes from South America and suggested that these patterns could be found for most Neotropical freshwater fishes. The phylogenetic analysis showed that both species *T.Kashmirensis* form separate clades, though it becomes evident that these are too close to each other. Furthermore, both phylogenetic tree shows that they are more close to *Triplophysastoliczkae* genus.

Kullander et al., (1999), places these species into Balitoridae family, but the phylogenetic relationship showed that these two species were more close to Nemacheilidae family than to Balitoridae. **Ward et al., 2005**, Muscle was harvested from a fresh specimen each collected from two different river systems, and was preserved in absolute ethanol. DNA was extracted using conventional phenol-chloroform method and re-suspended in nuclease free water. Polymerase chain reaction was performed to amplify mitochondrial cytochrome oxidase submit gene, using the forward primer fish R1. **Edgar 2004**, Gene Bank accession numbers for sequences used for the analysis are provided in Table 3 and Fig 1. Sequences were aligned using MUSCLE.

Nylander 2004, A Maximum Likelihood (ML) phylogeny was built in PHYML ver 3.0 (Guindon et al., 2010) using the dataset after finding out the best fit nucleotide substitution model using MrAIC. **Arunachalam et al.**, A genetic distance of 0.1% was observed between the *cox1* sequence of *H.thomassi*, from the Chalakkudy River and Kallada River, Interestingly, the sequence generated from the specimen collected from kallada River demonstrates the uniqueness with a sequence from the GenBank which was deposited – also in the Kallada River system) **Arunachalam et al., (2012)**, These three sequences formed a monophyletic group which was sister to the topotypic sequences of *H.lithopidos* collected for the present study. The average genetic distance between the sequence of topotypic *H.lithopidos*, collected from South Canada, and the *H.thomassi* sequences were 4.85%. This confirms that the sequence HMO1)715, collected from Rosemalai, is in fact. **See Ali et al., 2013**, Destructive fishing practices especially dynamiting is a major threat to the species in the streams around Pooyamkutty in Periyar River, as well as in Umayar, Rosemalai and Katalapara regions of Kalladariver. Poisoning and electric fishing are prevalent in the Sullia region of Dakshinakannada, especially when the water levels are low.

Devi & Ali 2011, Currently, *H. thomassi* has been listed as ‘Critically Endangered’ in the IUCN Red list of Threatened species based on limited distribution information available during the assessment. Additional information on distribution, threats and taxonomic clarifications have led to scenario where the conservation status of the species needs to be re-assessed. **Ali et al., 2013**, In spite of research that began more than two hundred years ago, our knowledge on the diversity and distribution of freshwater fishes of Western Ghats is far from complete. Studies such as those reported in the present paper, as well as others. **Emmanuel et al., 2013; night et al., 2013a,b,c**), are addressing this gap in knowledge on species distribution, widely termed as the ‘Wallacean shortfall’ thereby facilitating the development and implementation of conservation policies and action in this exceptional region of freshwater biodiversity. **Valsecchi et al., 2005**, The mitochondrial control region was chosen for phylogeographic analysis owing to high nucleotide polymorphism in skate species.

Clement et al., 2000, DNA sequences for a partial section of this region were generated for all sampled individuals. To reconstruct genealogical relationship among haplotypes, a maximum parsimony haplotype network was constructed using TCS1.2.1. **Clarke & Warwick 2001**, To test for population structure among samples, Bayesian clustering of samples was conducted with and gene flow between clusters was examined by estimating individual admixture coefficients. For detailed methods see electronic supplementary material, appendix B. **Smith et al., 2008**, To place results in a broader phylogenetic context, samples from an additional 13 skate species were collected (electronic supplementary material, appendix C), and sequenced for partial fragments of the mitochondrial DNA control region and cytochrome b gene. The cytochrome b gene species. Sequences were aligned alongside those of outgroups.

Nikula et al., 2007, Estimates of time from divergence in skate correspond to a period of cooling in the Arctic and Atlantic Oceans approximately 4 Ma, which has been hypothesized as contributing to divergence between species of *Macoma* bivalves. **Corrigan & Beheregaray 2009**, In the common skate, divergence could have been related to onset of cooler conditions emerging in the Arctic at that time, similar to suggestions that divergence in wobbegong sharks in Australia was ecologically driven and associated with the occupation of novel habitats. **Gomez et al., 2007**, An alternative explanation is that divergence was related to isolation within glacial refugia following the onset of Pleistocene ice sheet expansion as has been suggested for cryptic bryozoan species in the northeast Atlantic. **Ebert & Compagno 2007**, The conserved body morphology of skates, and their restriction to benthic habitats, belies a group with a high degree of biodiversity and endemism.

Tinti et al., 2003; Turan 2008, The phylogeny of northern European species in this study extends that of previous work demonstrating mono-phyly of tribes and genera within European Rajiformes. **Bickford et al., 2007**, Detailed morphological study of samples collected over wider spatial scales may help to uncover diagnostic features that separate these genetically segregated species. Until this taxonomic uncertainty is resolved, we regard these taxa as representing 'cryptic' species. Who consider two or more species to be cryptic if

they are, or have been, classified as a single nominal species because they are at least superficially morphologically indistinguishable. **Ahone et al., 2009**, Using similar methods, cryptic species have also been identified in hound sharks scalloped hammerhead sharks and wobbegong sharks and these methods have helped to confirm the taxonomic distinctness of Australian river sharks. They have also helped to identify strong spatial genetic grey nurse sharks.

Murta, 2000; Costa et al., 2003, The phenotypic discreteness suggests a direct relationship between the extent of phenotypic divergence and geographic separation, which indicated that geographic separation, is a limiting factor to migration among stocks. Both morphometric and meristic characters are important in morphological studies. Here in the current study all 22 morphometric characters were significantly different in the populations. Studies already showed that morphometric characters are more suitable than meristic characters to describe the intra specific variation. **Avise et al., 1987a; Moritz et al., 1987**, In animal eukaryotic cells, a small amount of DNA is found outside the nucleus within organelles in the cytoplasm. Mitochondria each contain about 10 identical copies of the mitochondrial genome. The genome is comparatively small (16000 to 19 000 nucleotides), double-stranded, and circular. The genome is contains specify essential subunits of components of the respiratory pathway as well as the transfer and ribosomal RNA needed to manufacture those protein subunits.

Vawter and Brown (1986) and Palumbi and Wilson (1990), both measured intraspecific mtDNA variation in the North American Sea urchin, *Strongylocentrotus purpuratus*. Their estimates of intraspecific mtDNA sequence divergence were 0.0099 and 0.0048 respectively. They did find subdivision between *S. droebachensis* sampled from eastern and western North America. The mean intraspecific divergence for this species, including samples from both coasts, was 0.001. **Powers et al., 1986**, The mtDNA of the deep-sea scallop (*Placopecten magellanicus*) is 34000 nucleotides long, about twice the metazoan average (Snyder et al., 1987). This scallop genome appears to be at the upper limit of size for non-yeast eukaryotic mtDNA, a position also occupied by the 26000 nucleotide mtDNA of the parasitic nematode *Romanomermis culicivorax*. **Edwards and Skibinski 1987** reported no size

heteroplasmy in mitochondrial genomes from mussels. They sampled 187 individuals from five localities in south-western England where these species are known to hybridize. Using three six-nucleotide restriction enzymes known to recognize polymorphic sites, they found 24 mtDNA haplotypes. The frequency of the four most common haplotypes was significantly different in four out of seven χ^2 tests between localities. Although no more sophisticated tests of subdivision were performed with the data, the study appears to show that mussel haplotypes are geographically subdivided along the coastline south-western England.

Bentzen et al., 1988, although a reanalysis of the data (Roff and Bentzen 1989) has shown them to be so. American eels and European eels spawn in the western tropical mid-Atlantic Ocean. Juveniles are transported throughout the North Atlantic by Ocean currents. **Avise et al., (1986)** assayed intraspecific mtDNA sequence variation in non-spawning adults collected from the western coast of North America and from England and Ireland. No evidence was found for genetic subdivision in either species. The mtDNA of the two species was shown to be different with 11 of the 14 restriction enzymes used. **O' Brien and Evermann 1988**, The marine species with low levels of mtDNA sequence diversity are a disparate assemblage, ranging from North American inshore sea urchins to anadromous fish. The cause of bottleneck events in these species could be biogeographical or epidemiological. **Peres-Neto and Magnan, 2004**, Meristic characters were considered to be fixed in early embryonic life of the individual and remain unchanged thereafter. Thus, they respond to environmental factors only a short period of time during embryonic development. In contrast, morphometric characters varied according to the changing environmental conditions of the habitat throughout their life and the phenotypic plasticity were shown in many freshwater fish species.

Piska (1990) and Serajuddin (2004), however, in some studies meristic characters were successfully used for stock identification and in accordance with this finding the 4 meristic characters in the present study were found to be highly significant. Being based on the morphometric and meristic variations the finding of the present study are similar to the finding of Chatterji et al., (1977a). **Cakmak and Alp (2010)**, reported morphological differences among the

Mesopotamian spiny eel *Mastacembelus mastacemelus* populations. The significant divergence in morphological and meristic traits revealed that intraspecific variation existed among the specimens of *C. lalia*. Therefore, multiracial composition of *C. lalia* across the environmental gradient cannot be ruled out. **Williams et al., 1990**, The development of molecular genetic technique has made possible to decipher the genetic constituent of fish that has been revolutionized our ability to understand the genetic divergence at subspecies level. Nowadays molecular markers are used in order to discriminate genetic diversity. The random amplified polymorphic DNA (RAPD) analysis is a technique based on the polymerase chain reaction (PCR), used to detect DNA polymorphism, by amplifying discrete regions of genome with short oligonucleotide primers of arbitrary sequence.

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Lehmann et al., (2000), beside this the technique was also used to study the phylogeny of freshwater fishes such as sturgeon by Comincini et al., (1998) and Anguillidae. **Barman et al., 2003**), RAPD was used to evaluate genetic diversity in various groups of fishes like guppy, catfish common carp (Bartfi et al., 2003) and Indian major carps. **Larson and Wilson, (1989)**, The nucleotide and phylogenetic relationship among the four fishes of gourami (Belontidae family) on the basis of D9 and D11 domain of 28S rDNA was carried out for the first time. 28S r RNA genes have also been used to study the evolutionary relationship between lungfishes, Coelacanth, rainbow trout, eel and sturgeon and tetrapods and thid id primarily due the fact these divergent domains differ extensively among distantly related taxa. **Hassouna et al., (1984)**, The D9 and D11 domain nucleotide sequences of Colisalalia ranged from 224bp to 337bp and 585bp to 613bp respectively. Divergent domains of 28S r RNA genes have been studied widely earlier in mouse and humans and 28S r RNA genes is due to some unique sities embedded within the largely conserved secondary structure.

Verma and Serajuddin (2012), this is first report phylogenetic analysis in four gourami species on the basis of divergent domains of 28S r RNA. However, also reported the divergent domain D8 of 28S r RNA gene in four species of fishes of family siluridae. **Reeb and Avise (1990)**, Sequence variation in the mitochondrial genome has been found to be geographically subdivided in some marine species. These include marine invertebrates such as sea urchins, mussels and hrsehoe crabs and toadfish have also been found to be genetically subdivided. **Avise et al., 1989**), Mean mtDNA sequence diversity is

the average nucleotide diversity between each pair of individual mitochondrial genomes in a population. Estimates of its vary by about three orders of magnitude for marine species and encompass the entire range of terrestrial intraspecific diversity.

Nei (1987), there are several explanations for the low amount of mtDNA sequence diversity observed in marine species. Mitochondrial genome evolution may be generally slower in some marine animals. Variation in the rate of genome evolution, both mitochondrial and nuclear, has often been reported. **Erlich (1989)**, on the horizon of marine population genetics is a new DNA analysis tool, the polymerase chain reaction. This innovative method could be used to produce large quantities of mitochondrial or nuclear DNA from marine individuals whose small body size or state of preservation precludes the use of present DNA analysis techniques. The newly synthesized double-stranded DNA is then denatured by heat into single strands, and the cycle is repeated. Each cycle results in a 100% increase in the amount of target DNA so that, after 20-30 cycles, the target DNA predominates in the reaction mixture.

Satta et al., (1988), Animal mtDNA is thought to be strictly maternally inherited. The results of at least two experimental tests are consistent with this; one study used the tobacco budworm and the other used freshwater fish, In both cases, hybrids were produced between two strains having different mtDNA. This case for maternal mtDNA inheritance must remain open, however, as inheritance patterns amongst some *Drosophila* hybrids can be explained by a paternal mtDNA contribution. **Palumbi and Wilson (1990)**, despite the assumed homogenizing effect of the ocean, complete barriers to reproduction do exist in the marine environment as they are a prerequisite for speciation. The perceived nature of marine reproductive barriers is limited by our terrestrial imaginations. Some obvious examples include the formation of land bridges by falling sea levels or rising continents. **Hadwen, Arthinton&Mosisch, 2003**, Even well-protected conservation areas can become focal points for tourism and recreational activities that may reduce habitat quality and biodiversity.

Leveque &Balian (2005), If science is to be deployed in a manner that will secure commitment to the conservation of freshwater biodiversity from

politicians and decision-makers, scientists will have to make some adjustments in attitude. **Adams et al., 2004**, an apparent lack of common ground between organisations committed to sustaining livelihoods and those concerned with biodiversity conservation might arise from different starting points and prioritisation but they need not imply that attempts to combine the goals of conservation and meeting human needs should be regarded as fertile. **Stearns & Stearns. Meffe, 2001**, Pragmatic approaches will be needed to ensure that attempts at communication between freshwater scientists and water resource managers, as well as other stakeholders, contribute to planning and problem solving and do not become a dialogue of the deaf. This is a significant challenge as motivations of the broader community may be neither open nor fair. Conservation typically operates in a world where many players are characterized by dishonesty, self-interest, and hostility to nature, and where corporate interests often assume disproportionate importance.

Edwards 2009, In addition, the rapid evolution of mtDNA, while providing resolution to shallow branches of a phylogenetic tree, may result in poorly resolved deeper nodes due to saturation. As described above, the use of nuclear loci were precluded by the polyploidy observed in many cyprine taxa making assessment of homology difficult. Future research on this group should include the development of nuclear loci and resolution of issues associated with polyploidy to construct a multi-locus species tree that may better represent the genealogical relationship of the taxa in this clade. **Funk & Omland 2003**, Molecular phylogenetic analysis of the taxa sampled in this study provides insight into the sister group relationship of this relatively unknown group of Cypriniformes and significantly aids in clarification of taxonomic and classification issues that have long been unresolved. Our conclusions are tempered by the fact that the phylogenetic hypothesis supported by our analysis is based solely on mitochondrial genes and thus represents data from what is essentially a single locus.

POSTLETHWAIT et al., 2000, The duplications of the LWS/MWS genes. Form another tandem gene cluster. These results suggest that all counterparts of the opsin gene pairs by the whole genome duplication have been lost or become pseudogenes by mutations, as in the case of many other genes in

the zebrafish genome. **Clement et al., 2000**, DNA sequences for a partial section of this region were generated for all sampled individuals (GeneBank accessions GQ392065 to GQ3920981, electronic supplementary material, appendix A and for detailed methods, appendix B) to reconstruct genealogical relationship among haplotypes, a maximum parsimony haplotype network was constructed using TCS 1.2.1. **Delsuc et al., 2005** the emerging field of phylogenomics is a result of this explosive data boom, representing the study of complete genomes from many closely related species using an evolutionary perspective.

Avise 2004, Protein electrophoresis improved on album immunology by providing a cost-effective approach to surveying genetic variation across multiple nuclear markers and resulted advances in population genetic theory because allozyme data assay Mendelian markers. Many allozyme studies have employed phylogenetic and population genetic analysis to infer both intra and interspecific evolutionary and biogeographical histories in animals, plants, and fungi.

Shepherd et al., 2005, an alternative, but powerful, approach is one that uses serial genetic data such as historic or ancient DNA within a single species. By using serial sets of fossil genetic data within a locality it is possible to reconstruct the history of a population through time. **Bergsten et al., 2012**, The present data (with 98% species coverage) comprise the first comprehensive molecular study on the freshwater fish diversity of a complete BH and will serve as reference for future studies of this large fauna. Our finding that model-based clustering of sequences outperformed the simplest, but most widely used version of distance based clustering is consistent with theoretical and previous empirical work.

Allendorf et al., 1987; Wimberger, 1992, In general, fish demonstrate greater variances in morphological traits both within and between populations than any other vertebrates, and are more susceptible to environmentally induced morphological variations. **Murta, 2000; Costa et al., 2003**, The phenotypic discreteness suggests a direct relationship between the extent of phenotypic divergence and geographic separation, which indicated that geographic

separation, is a limiting factor to migration among stocks. Both morphometric and meristic characters are important in morphological studies. Here in the current study all 22 morphometric characters were significantly different in the populations. Studies already showed that morphometric characters are more suitable than meristic characters to describe the intra specific variation. **Hinder and Johnson, 1993; Peres-Neto and Magnan, 2004**, meristic characters were considered to be fixed in early embryonic life of the individual and remain unchanged thereafter. Thus, they respond to environmental factors only a short period of time during embryonic development. In contrast, morphometric characters varied according to the changing environmental conditions of the habitat throughout their life and the phenotypic plasticity were shown in many freshwater fish species. **Pakkasamaa et al., (1998)**, also reported the pronounced interspecific variation as compared to intraspecific variations in four land locked salmonid species. **Cakmak and Alp (2010)**, reported morphological differences among the Mesopotamian spiny eel *Mastacembelus mastacemelus* populations. The significant divergence in morphological and meristic traits revealed that intraspecific variation existed among the specimens of *C. lalia*. Therefore, multiracial composition of *C. lalia* across the environmental gradient cannot be ruled out. **Welsh and McClelland, 1990; Williams et al., 1990**, The development of molecular genetic technique has made possible to decipher the genetic constituent of fish that has been revolutionized our ability to understand the genetic divergence at subspecies level. Nowadays molecular markers are used in order to discriminate genetic diversity. The randomly amplified polymorphic DNA (RAPD) analysis is a technique based on the polymerase chain reaction (PCR), used to detect DNA polymorphism, by amplifying discrete regions of genome with short oligonucleotide primers of arbitrary sequence. **Garg et al., 2009a, 2009b**, This technique was used extensively to detect genetic diversity in animals (Cushwa and Medrano, 1996).

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Pahgiofilinaria (60%) .**Larson and Wilson, 1989**, The nucleotide composition and phylogenetic relationship among the four fishes of gourami (Belontiidae family) on the basis of D9 and D11 domain of 28S r DNA was carried out for the first time. 28S r RNA genes have also been used to study the evolutionary relationship between lungfishes, Coelacanth, rainbow trout, eel and sturgeon and tetrapods (Zordoya et al., 1996) and this is primarily due the fact these divergent domains differ extensively among distantly related taxa . **Verma and Serajuddin (2012)**, this is first report of phylogenetic analysis in four gourami species on the basis of divergent domains of 28S r RNA. However, also reported the divergent domain D8 of 28S r RNA gene in four species of fishes of family siluridae.

Avise 1996a, The field of conservation genetics traditionally has been associated with concerns about levels of genetic variation and inbreeding depression in small, often captive animal populations. **Norse (1994)**, Lamented that “Marine conservation biology lags terrestrial conservation biology by about 20 years. It lacks central paradigms, graduate training programs, and substantial dedicated funding.” On the other hand, the marine realm has provided some of the most exciting, imaginative, and innovative of available case studies in all of conservation genetics. I hope this issue may contribute to a wider appreciation of such efforts and also elp to enthuse a new generation of widely versed biologists about this empirically and conceptually rich and important field.**Biol. Trop 2016**, In the present study, the numbers of morphometric measurements used are considered to be of adequate number for identification and differentiation of these two fish species. Compared with the earlier reports of Kulander et al., (1991) and Talwar and Jhingran (1991) who reported only significant variation in MBD, SLL and MinBD between the species, this study reported significant variation in 15 morphometric characters than can be used in their identification and differentiation. For most of the morphometric was less evident, and this suggests that each species consists of a phenotypically and genetically homogeneous group. **Jiuxuan, Kun, Shengjuan, Xiuyue&Zhaobin (2015)**, our study provides the first molecular characterization of these species from Kashmir valley. The overall base composition of both. The average K2P divergence within conspecific specimens was found only 0.001 and among

congeneric species 0.007. **Ward, Hanner and Hebert (2009), Hubert et al., (2008)** found average K2P values of 0.27% (conspecific) and 8.37% (congeneric), discriminating 93% of the species while studying freshwater fishes from Canada. In another study 93% of freshwater fishes were discriminated from Mexico and Guatemala showing average K2P values of 0.45% in conspecific and 5.1% in congeneric (Valdez-Moreno, Ivanova, Elias-Guitierrez. **Hubert et al., (2007)**, similarly pointed to a recent radiation of some groups of freshwater fishes from South America and suggested that these patterns could be found for most Neotropical freshwater fishes. The phylogenetic analysis showed that both species *T.Kashmirensis* form separate clades, though it becomes evident that these are too close to each other. Furthermore, both phylogenetic tree shows that they are more close to *Triplophysastoliczkae* genus. **Kullander et al., (1999)**, places these species into Balitoridae family, but the phylogenetic relationship showed that these two species were more close to Nemacheilidae family than to Balitoridae. **Ward et al., 2005**, Muscle was harvested from a fresh specimen each collected from two different river systems, and was preserved in absolute ethanol. DNA was extracted using conventional phenol-chloroform method and re-suspended in nuclease free water. Polymerase chain reaction was performed to amplify mitochondrial cytochrome oxidase submit gene, using the forward primer fish R1. **Edgar 2004**, Gene Bank accession numbers for sequences used for the analysis are provided in Table 3 and Fig 1. Sequences were aligned using MUSCLE. **Nylander 2004**, A Maximum Likelihood (ML) phylogeny was built in PHYML ver 3.0 (Guindon et al., 2010) using the dataset after finding out the best fit nucleotide substitution model using MrAIC. **Arunachalam et al.**, A genetic distance of 0.1% was observed between the *cox1* sequence of *H.thomassi*, from the Chalakkudy River and Kallada River, Interestingly, the sequence generated from the specimen collected from kallada River demonstrates the uniqueness with a sequence from the GenBank which was deposited – also in the Kallada River system) **Arunachalam et al., (2012)**, These three sequences formed a monophyletic group which was sister to the topotypic sequences of *H.lithopidos* collected for the present study. The average genetic distance between the sequence of topotypic *H.lithopidos*, collected from South Canada, and the *H.thomassi*

sequences were 4.85%. This confirms that the sequence HMO!)715, collected from Rosemalai, is in fact. **See Ali et al., 2013**, Destructive fishing practices especially dynamiting is a major threat to the species in the streams around Pooyamkutti in Periyar River, as well as in Umayar, Rosemalai and Katalapara regions of Kalladriver. Poisoning and electric fishing are prevalent in the Sullia region of Dakshinakannada, especially when the water levels are low. **Devi & Ali 2011**, Currently, *H. thomasi* has been listed as 'Critically Endangered' in the IUCN Red list of Threatened species based on limited distribution information available during the assessment. Additional information on distribution, threats and taxonomic clarifications have led to scenario where the conservation status of the species needs to be re-assessed. **Ali et al., 2013**, In spite of research that began more than two hundred years ago, our knowledge on the diversity and distribution of freshwater fishes of Western Ghats is far from complete. Studies such as those reported in the present paper, as well as others. **Emmanuel et al., 2013; night et al., 2013a,b,c**), are addressing this gap in knowledge on species distribution, widely termed as the 'Wallacean shortfall' thereby facilitating the development and implementation of conservation policies and action in this exceptional region of freshwater biodiversity. **Valsecchi et al., 2005**, The mitochondrial control region was chosen for phylogeographic analysis owing to high nucleotide polymorphism in skate species. **Clement et al., 2000**, DNA sequences for a partial section of this region were generated for all sampled individuals. To reconstruct genealogical relationship among haplotypes, a maximum parsimony haplotype network was constructed using TCS1.2.1. **Clarke & Warwick 2001**, To test for population structure among samples, Bayesian clustering of samples was conducted with and gene flow between clusters was examined by estimating individual admixture coefficients. For detailed methods see electronic supplementary material, appendix B. **Smith et al., 2008**, To place results in a broader phylogenetic context, from an additional 13 skate species were collected (electronic supplementary material, appendix C), and sequenced for partial fragments of the mitochondrial DNA control region and cytochrome b gene. The cytochrome b gene species. Sequences were aligned alongside those of outgroups. **Nikula et al., 2007**, Estimates of time from divergence in skate

correspond to a period of cooling in the Arctic and Atlantic Oceans approximately 4 Ma, which has been hypothesized as contributing to divergence between species of *Macoma* bivalves. **Corrigan & Beheregaray 2009**, In the common skate, divergence could have been related to onset of cooler conditions emerging in the Arctic at that time, similar to suggestions that divergence in wobbegong sharks in Australia was ecologically driven and associated with the occupation of novel habitats. **Gomez et al., 2007**, An alternative explanation is that divergence was related to isolation within glacial refugia following the onset of Pleistocene ice sheet expansion as has been suggested for cryptic bryozoan species in the northeast Atlantic. **Ebert & Compagno 2007**, The conserved body morphology of skates, and their restriction to benthic habitats, belies a group with a high degree of biodiversity and endemism. **Tinti et al., 2003; Turan 2008**, The phylogeny of northern European species in this study extends that of previous work demonstrating monophyly of tribes and genera within European Rajiformes. **Bickford et al., 2007**, Detailed morphological study of samples collected over wider spatial scales may help to uncover diagnostic features that separate these genetically segregated species. Until this taxonomic uncertainty is resolved, we regard these taxa as representing 'cryptic' species. Who consider two or more species to be cryptic if they are, or have been, classified as a single nominal species because they are at least superficially morphologically indistinguishable. **Ahne et al., 2009**, Using similar methods, cryptic species have also been identified in hound sharks, scalloped hammerhead sharks and wobbegong sharks and these methods have helped to confirm the taxonomic distinctness of Australian river sharks. They have also helped to identify strong spatial genetic structure in grey nurse sharks. **Murta, 2000; Costa et al., 2003**, The phenotypic discreteness suggests a direct relationship between the extent of phenotypic divergence and geographic separation, which indicated that geographic separation, is a limiting factor to migration among stocks. Both morphometric and meristic characters are important in morphological studies. Here in the current study all 22 morphometric characters were significantly different in the populations. Studies already showed that morphometric characters are more suitable than meristic characters to describe the intra-specific variation. **Avise et al., 1987a; Moritz et al., 1987**, In animal eukaryotic

cells, a small amount of DNA is found outside the nucleus within organelles in the cytoplasm. Mitochondria each contain about 10 identical copies of the mitochondrial genome. The genome is comparatively small (16000 to 19 000 nucleotides), double-stranded, and circular. The genome contains specific essential subunits of components of the respiratory pathway as well as the transfer and ribosomal RNA needed to manufacture those protein subunits.

Vawter and Brown (1986) and Palumbi and Wilson (1990), both measured intraspecific mtDNA variation in the North American Sea urchin, *Strongylocentrotus purpuratus*. Their estimates of intraspecific mtDNA sequence divergence were 0.0099 and 0.0048 respectively. They did find subdivision between *S. droebachensis* sampled from eastern and western North America. The mean intraspecific divergence for this species, including samples from both coasts, was 0.001.

Powers et al., 1986, The mtDNA of the deep-sea scallop (*Placopecten magellanicus*) is 34000 nucleotides long, about twice the metazoan average (Snyder et al., 1987). This scallop genome appears to be at the upper limit of size for non-yeast eukaryotic mtDNA, a position also occupied by the 26000 nucleotide mtDNA of the parasitic nematode *Romanomermis culicivorax*.

Edwards and Skibinski 1987 reported no size heteroplasmy in mitochondrial genomes from mussels. They sampled 187 individuals from five localities in south-western England where these species are known to hybridize. Using three six-nucleotide restriction enzymes known to recognize polymorphic sites, they found 24 mtDNA haplotypes. The frequency of the four most common haplotypes was significantly different in four out of seven χ^2 tests between localities. Although no more sophisticated tests of subdivision were performed with the data, the study appears to show that mussel haplotypes are geographically subdivided along the coastline south-western England.

Bentzen et al., 1988, although a reanalysis of the data (Roff and Bentzen 1989) has shown them to be so. American eels and European eels spawn in the western tropical mid-Atlantic Ocean. Juveniles are transported throughout the North Atlantic by Ocean currents. **Avise et al., (1986)** assayed intraspecific mtDNA sequence variation in non-spawning adults collected from the western coast of North America and from England and Ireland. No evidence was found for genetic subdivision in either species. The mtDNA of the two

species was shown to be different with 11 of the 14 restriction enzymes used. **O'Brien and Evermann 1988**, The marine species with low levels of mtDNA sequence diversity are a disparate assemblage, ranging from North American inshore sea urchins to anadromous fish. The cause of bottleneck events in these species could be biogeographical or epidemiological. **Peres-Neto and Magnan, 2004**, Meristic characters were considered to be fixed in early embryonic life of the individual and remain unchanged thereafter. Thus, they respond to environmental factors only a short period of time during embryonic development. In contrast, morphometric characters varied according to the changing environmental conditions of the habitat throughout their life and the phenotypic plasticity were shown in many freshwater fish species. **Piska (1990) and Serajuddin (2004)**, however, in some studies meristic characters were successfully used for stock identification and in accordance with this finding the 4 meristic characters in the present study were found to be highly significant. Being based on the morphometric and meristic variations the finding of the present study are similar to the finding of Chatterji et al., (1977a). **Cakmak and Alp (2010)**, reported morphological differences among the Mesopotamian spiny eel *Mastacembelus mastacemelus* populations. The significant divergence in morphological and meristic traits revealed that intraspecific variation existed among the specimens of *C. lalia*. Therefore, multiracial composition of *C. lalia* across the environmental gradient cannot be ruled out. **Williams et al., 1990**, The development of molecular genetic technique has made possible to decipher the genetic constituent of fish that has been revolutionized our ability to understand the genetic divergence at subspecies level. Nowadays molecular markers are used in order to discriminate genetic diversity. The random amplified polymorphic DNA (RAPD) analysis is a technique based on the polymerase chain reaction (PCR), used to detect DNA polymorphism, by amplifying discrete regions of genome with short oligonucleotide primers of arbitrary sequence. **Garg et al., 2009a, 2000b**, this technique was used extensively to detect genetic diversity in animals (Cushwa and Medrano, 1996). RAPD technique is considered to be one of the best techniques for taxonomic systematic study of various organisms. **Ertas and Seker, 2005**, RAPD markers have many applications like gene mapping,

population genetics, molecular evolutionary genetics and plant and animal breeding. **Apostolodis et al., (2011)**, RAPD technique was found to be more sensitive than mitochondrial DNA analysis which failed to detect DNA polymorphism in the populations of tilapia (Capilai, 1990; Seyoum and Komfield, 1992) while Bardakei and Skibinski (1994) reported intraspecific variation in tilapia using RAPD technique. Several authors used RAPD as a tool for the assessment of genetic variability reported the genetic diversity in the Balkan trout population (*Salmo trutta*). RAPD tool was also used in molecular typing of bacterial strains which affect the fish population. **Castro et al., (2012)**, In the same way also reported genetic variability of bacterial strain, *Edwardsiella ictaluri* sampled from the fish namely Senegalese sole, *Solea senegalensis* (Kaup) from Spain. The study suggested that RAPD markers can be used for estimation of genetic variation and these genetic variation data can be used for stock selection for selective breeding programmes. **Welsh et al., (1991)**, in the present study, the analysis of RAPD data screened genetic variations among the populations of *C. lalia*. The number of amplified bands ranged between 282-420 with size ranging from 150 bp-2500 bp. Actually, the number and size of band generated strictly depends upon primer sequence and the DNA template. **Wei et al., 2006**, Different levels of genetic variability were detected among the three populations of the *C. lalia*. The percentage of polymorphic loci and heterozygosity for the population are considered as indicators of the actual level of genetic variability of the species. A low level of intra-specific polymorphism in terms of the proportion of polymorphic (55.60%) loci was detected in the current study by the RAPD markers. Similar low rate of polymorphism was detected. **Lehmann et al., (2000)**, besides this the technique was also used to study the phylogeny of freshwater fishes such as sturgeon by Comincini et al., (1998) and Anguillidae. **Barman et al., 2003**, RAPD was used to evaluate genetic diversity in various groups of fishes like guppy, catfish common carp (Bartfi et al., 2003) and Indian major carps. **Larson and Wilson, (1989)**, the nucleotide and phylogenetic relationship among the four fishes of gourami (*Belontiidae* family) on the basis of D9 and D11 domain of 28S rDNA was carried out for the first time. 28S rRNA genes have also been used to study the evolutionary relationship between lungfishes, Coelacanth, rainbow trout, eel

and sturgeon and tetrapods and this is primarily due to the fact these divergent domains differ extensively among distantly related taxa. **Hassouna et al., (1984)**, The D9 and D11 domain nucleotide sequences of *Colisalapia* ranged from 224bp to 337bp and 585bp to 613bp respectively. Divergent domains of 28S rRNA genes have been studied widely earlier in mouse and humans and 28S rRNA genes is due to some unique sites embedded within the largely conserved secondary structure. **Verma and Serajuddin (2012)**, This is first report phylogenetic analysis in four gourami species on the basis of divergent domains of 28S rRNA. However, also reported the divergent domain D8 of 28S rRNA gene in four species of fishes of family siluridae. **Reeb and Avise (1990)**, Sequence variation in the mitochondrial genome has been found to be geographically subdivided in some marine species. These include marine invertebrates such as sea urchins, mussels and horseshoe crabs and toadfish have also been found to be genetically subdivided. **Avise et al., (1989)**, Mean mtDNA sequence diversity is the average nucleotide diversity between each pair of individual mitochondrial genomes in a population. Estimates of it vary by about three orders of magnitude for marine species and encompass the entire range of terrestrial intraspecific diversity. **Nei (1987)**, there are several explanations for the low amount of mtDNA sequence diversity observed in marine species. Mitochondrial genome evolution may be generally slower in some marine animals. Variation in the rate of genome evolution, both mitochondrial and nuclear, has often been reported. **Erlich (1989)**, On the horizon of marine population genetics is a new DNA analysis tool, the polymerase chain reaction. This innovative method could be used to produce large quantities of mitochondrial or nuclear DNA from marine individuals whose small body size or state of preservation precludes the use of present DNA analysis techniques. The newly synthesized double-stranded DNA is then denatured by heat into single strands, and the cycle is repeated. Each cycle results in a 100% increase in the amount of target DNA so that, after 20-30 cycles, the target DNA predominates in the reaction mixture. **Satta et al., (1988)**, Animal mtDNA is thought to be strictly maternally inherited. The results of at least two experimental tests are consistent with this; one study used the tobacco budworm and the other used freshwater fish. In both cases, hybrids

were produced between two strains having different mtDNA. These cases for maternal mtDNA inheritance must remain open, however, as inheritance patterns amongst some *Drosophila* hybrids can be explained by a paternal mtDNA contribution. **Palumbi and Wilson (1990)**, despite the assumed homogenizing effect of the ocean, complete barriers to reproduction do exist in the marine environment as they are a prerequisite for speciation. The perceived nature of marine reproductive barriers is limited by our terrestrial imaginations. Some obvious examples include the formation of land bridges by falling sea levels or rising continents. **Hadwen, Arthinton&Mosisch, 2003**, Even well-protected conservation areas can become focal points for tourism and recreational activities that may reduce habitat quality and biodiversity. **Leveque &Balian (2005)**, If science is to be deployed in a manner that will secure commitment to the conservation of freshwater biodiversity from politicians and decision-makers, scientists will have to make some adjustments in attitude. **Adams et al., 2004**, an apparent lack of common ground between organizations committed to sustaining livelihoods and those concerned with biodiversity conservation might arise from different starting points and prioritisation but they need not imply that attempts to combine the goals of conservation and meeting human needs should be regarded as fertile. **Stearns & Stearns. Meffe, 2001**, Pragmatic approaches will be needed to ensure that attempts at communication between freshwater scientists and water resource managers, as well as other stakeholders, contribute to planning and problem solving and do not become a dialogue of the deaf. This is a significant challenge as motivations of the broader community may be neither open nor fair. Conservation typically operates in a world where many players are characterized by dishonesty, self-interest, and hostility to nature, and where corporate interests often assume disproportionate importance. **Edwards 2009**, In addition, the rapid evolution of mtDNA, while providing resolution to shallow branches of a phylogenetic tree, may result in poorly resolved deeper nodes due to saturation. As described above, the use of nuclear loci were precluded by the polyploidy observed in many cyprine taxa makin assessment of homology difficult. Future research on this group should include the development of nuclear loci and resolution of issues associated with polyploidy to construct a multi-locus species tree that may better represent the

genealogical relationship of the taxa in this clade. **Funk & Omland 2003**, Molecular phylogenetic analysis of the taxa sampled in this study provides insight into the sister group relationship of this relatively unknown group of Cypriniformes and significantly aids in clarification of taxonomic and classification issues that have long been unresolved. Our conclusions are tempered by the fact that the phylogenetic hypothesis supported by our analysis is based solely on mitochondrial genes and thus represents data from what is essentially a single locus. **Postlethwait et al., 2000**, The duplications of the LWS/MWS genes. Form another tandem gene cluster. These results suggest that all counterparts of the opsin gene pairs by the whole genome duplication have been lost or become pseudo genes by mutations, as in the case of many other genes in the zebra fish genome. **Clement et al., 2000**, DNA sequences for a partial section of this region were generated for all sampled individuals (GeneBank accessions GQ392065 to GQ3920981, electronic supplementary material, appendix A and for detailed methods, appendix B) To reconstruct genealogical relationship among haplotypes, a maximum parsimony haplotype network was constructed using TCS 1.2.1. **Delsuc et al., 2005** the emerging field of phylogenomics is a result of this explosive data boom, representing the study of complete genomes from many closely related species using an evolutionary perspective. **Avise 2004**, Protein electrophoresis improved on album immunology by providing a cost-effective approach to surveying genetic variation across multiple nuclear markers and resulted advances in population genetic theory because allozyme data assay Mendelian markers. Many allozyme studies have employed phylogenetic and population genetic analysis to infer both intra and interspecific evolutionary and biogeographical histories in animals, plants, and fungi. **Shepherd et al., 2005**, an alternative, but powerful, approach is one that uses serial genetic data such as historic or ancient DNA within a single species. By using serial sets of fossil genetic data within a locality it is possible to reconstruct the history of a population through time. **Bergsten et al., 2012**, The present data (with 98% species coverage) comprise the first comprehensive molecular study on the freshwater fish diversity of a complete BH and will serve as reference for future studies of this large fauna.

III - METHODOLOGY

MATERIALS AND METHODS

Ecological study

In total of 15 sites (5 low, 5 mid and 5 high elevations) will be selected from Palani hills. In each site, three replicas will be made between 1 m intervals. Month-wise sampling will be carried out for a year. The physico-chemical parameters of sampling site will be measured by following: latitude, longitude and elevation of sampling sites will be taken from GPS (Global Positioning System); temperature, pH, dissolved oxygen, velocity depth, substrates, etc will be measured. The fish surveys and identification will be carried out through non-destructive sampling by using sampling methods suited to the nature of river course, stream order, flow, presence of aquatic vegetation and local human disturbance. Visual surveys, cast nets, gill-nets and hooks and lines were used to sample fish species. Fish species will be identified with the aid of taxonomic keys and field guides (Jayaram 1981, 2010). Information on threats to and ecology of species will also be collected based on observation, measurements of stream related ecological covariates and semistructured interviews with local key informants. The collected fishes will be preserved separately in the field using 99% ethanol.

Isolation of Fish from In this study 15 fish samples were collected from 15 different Streams in palani hills of the southern western ghats India. Fish samples collected were *Carasius* sp., *Nandus* sp., *Mystus* sp., *Barbus* sophore, *Barbus* sp., *Saccobranchnus fossilis*, *Barbus stigma* and *Chela bacaila*. Each fish sample was placed in aseptic small plastic box, labeled and sealed separately to avoid contamination. These fishes were brought to the laboratory and kept at 40C. In all whole fish specimens dry, make grind powder were taken for analysis. Each fish was swabbed from gills and body surface and inoculated in Nutrient Broth. Then incubation at 370C for 24hrs was given. The bacterial isolates were grown on solid medium in TSA and BHI plates then they were incubated at 370C for 24hrs. The plate yielded bacterial lawn. Culture was taken from different sides of plate and streaked on the TSA and BHI plate. Single colony was picked up from the plate and streaked on to the TSA and BHI plate.

Total 20 isolates were purified for further characterization. Morphological and Biochemical characterization each isolate was tested for morphology, motility and Gram Stain. Bacterial motility was observed directly on media having agar concentration of 0.4% or less. The biochemical tests for identification of *Pseudomonas* isolates were performed. For each isolates IMVIC; Indole test, Methyl red test, VogesProskauer test (VP test), Citrate utilization test, Catalase test Oxidase, Esculin and Gelatin Hydrolysis (GT test) tests were done.

Molecular Characterization Isolation of DNA The isolated colonies were cultured in Luria-bertani broth and incubated at 37°C for 48hrs. Two ml of bacterial culture was centrifuged at 6000 rpm for about 10 minutes. One ml of lysis buffer (10mM Tris HCl, pH 8.0; 0.5M EDTA; 0.5% SDS; 1M NaCl) was added to the pellet and vortexed properly and incubated at 45°C in boiling water bath for 10 mins. Then 1ml of phenol: chloroform mixture (1:1) was added to the mixture and centrifuged at 10,000 rpm for 10 minutes at 40°C. The upper aqueous layer was transferred to another eppendorf and equal volume of chloroform: isoamyl alcohol mixture (24:1) was added and then 1/10th volume of 3M sodium acetate was added. The contents are mixed properly and centrifuged at 10,000 rpm for 10 minutes at 40°C. To the upper aqueous layer double the volume of chilled ethanol was added to precipitate the DNA and later centrifuged at 12,000 rpm for 10 minutes at 40°C. The DNA pellet was stored in 20-50 µl of TE buffer and stored at 4°C for further use. Isolated DNA was then detected by Gel Electrophoresis (Fig 1).

PCR amplification PCR amplification was performed for 16S ribosomal DNA (rDNA) from cultured bacteria using the universal primers fD1 (5'-AGA GTT TGA TCC TGG CTC AG-3') and Rp2 (5'-ACG GCT ACC TTG TTA ACG ACT T-3'). It was done in 25 µl reaction mixture using PCR master mix (2X) supplied by Thermo Scientific. Template DNA (3 µl) was used. Amplifications were carried out for 35 cycles (94 °C for 30 s, 55 °C for 30 s, and 72 °C for 60 s) in an Eppendorf Master Cycler Personal with an initial denaturizing at 94 °C for 5 min and a final extension at 72 °C for 7 min. Amplicons were detected by electrophoresis on a 1.5 %.

3.1. COLLECTION OF SAMPLING:

A total of 15 specimens (12 individuals from each sampling sites with size ranging 5.4–8.1cm) of fish were collected from Guntar, kurusedi, Fairy falls, Silver cascade, Moolayar, Thalakkuttu falls, Adukkam, Pachalur, kumbakarai Poolathur, Rat tail falls, using cast and drag nets. These regions are geographically isolated and characterized by different environmental conditions. The focal point was elucidation of ecotype. The specimens were brought to the laboratory for their studies.

PALANI HILLS OF DESCRIPTION:

Palani Hills (Tamil:பழநி மலை) are a mountain range in the southern Indian states of Kerala and Tamil Nadu. The Palani Hills are an eastward extension of the Western Ghats ranges, which run parallel to the west coast of India. The Palani Hills adjoin the high Anamalai range (Kerala) on the west, and extend east into the plains of Tamil Nadu, covering an area of 2,068 square kilometres (798 sq mi). The highest part of the range is in the southwest, and reaches 1,800-2,500 metres (5,906-8,202 feet) elevation; the eastern extension of the range is made up of hills 1,000-1,500 m (3,281-4,921 ft) high.^[1] It is also home to the main temple of Lord Muruga, who is worshipped as the primary god in Tamil Nadu.

Vandaravu peak is the highest peak in the Palani hills. The range lies between the Cumbum Valley on the south, which is drained by the Vaigai River and its upper tributaries, and the Kongunadu region to the north. The northern slopes are drained by the Shanmukha River, Nanganji River, and Kodavanar River, which are tributaries of the Kaveri River. The range lies mostly within Dindigul district, except in the western portion, where it forms the boundary between Dindigul district and Theni district to the south and Idukki District to the south west. The hill station of Kodaikanal lies in the southern central portion of the range. Palani Hills joined with Anamalai Hills and Cardamom Hills at Anamudi peak in Kerala state.

The Palani Hills are currently subject to increasing development pressure as it is under developed for a long time. The Palani Hills Conservation Council, a non-governmental organization headquartered in Kodaikanal, was founded in

1985. In the early 1990s the Tamil Nadu Forest Department proposed to the Tamil Nadu state government that much of the range be granted protected status as a wildlife sanctuary or Palani Hills Wildlife Sanctuary and National Park.^[2]

The Palani Hills are a mountain range in the southern Indian states of Kerala and Tamil Nadu. The Palani Hills are an eastward extension of the Western Ghats ranges, which run parallel to the west coast of India. Wikipedia

Elevation: 2,533 m

Location: Tamil Nadu, India

Mountain range: Western Ghats

Country: India

Easiest route: Laws Ghat Road

Parent range: Western Ghats

Palni Hills, range of hills, an eastward extension of the Western Ghats, in southwestern Tamil Nadu state, southern India. The range is a continuation of the Anaimalai Hills in Kerala state. The Palnis are about 45 miles (70 km) wide and 15 miles (23 km) long. In the south the hills terminate abruptly in steep slopes. The upper Palnis, in the west, consist of rolling hills covered with coarse grasses; dense forests grow in the valleys. Peaks include Vandaravu, 8,376 feet (2,553 metres); Vembadi Shola, 8,221 feet (2,505 metres); and Karunmakadu, 8,042 feet (2,451 metres). The town of Kodaikanal is located in a high basin about 7,000 feet (2,150 metres) above sea level. Potatoes, beans, root crops, pears, and peaches are cultivated in and around the hill villages. There are also bauxite mines.

The lower Palnis, in the east, form a jumble of peaks that range in elevation from 3,000 to 5,000 feet (900 to 1,500 metres) and are separated by steep wooded valleys. Teak trees have been extensively planted. Important cash crops include coffee, bananas, cardamom, citrus fruit, and turmeric.

SOURCES: GOOGLE SEARCH -WIKIPEDIA

PALANI HILLS OF SOUTH INDIA



MAP OF PALANI HILLS



3.2. DNA ISOLATION:

For the isolation of total genomic DNA, phenol-chloroform method was used. Muscle tissue powder (100-200mg) was placed in 2 ml micro-centrifuge tube with 500 μ l lysis buffer and the tissue was homogenized properly by using eppendorf micro-pestle. In the homogenized tissue again 500 μ l of lysis buffer and 10% SDS were added and mix properly in vortex and then incubate it overnight at room temperature for complete lysis of cells. After incubation, chilled 100 μ l of NaCl (4 M), 10 μ l of proteinase K enzyme (10mg/ml) was added properly. After mixing incubated it for 1 hrs at room temperature. After NaCl and protein kinase treatment, phenol, chloroform and isoamyl alcohol was added (25:24:1 ratio) in that centrifuge tube, mixed uniformly and centrifuges at 12000 rpm at room temperature for 12 minutes. The supernatant was then transferred to new micro-centrifuge tube and phenol-chloroform was added again (4:1 ratio), then mixed properly and centrifuge tube and phenol-chloroform was added again (4:1 ratio), then mixed properly and centrifuged at 12000 rpm for 10 minutes. After centrifugation upper aqueous phase was transferred in 1.5 ml microcentrifuge tube and 100 μ l sodium acetate (3M, pH 5.2) and 1 ml of chilled absolute alcohol was added for precipitation of DNA. After 1 min precipitation DNA pellet was washed with 70% alcohol. The pellet

was centrifuged at 12000 rpm for 2 minutes. Then the DNA pellet was dried and dissolved in TE buffer. The amplified DNA fragments were separated on 1% agarose gel (Himedia) and stained with ethidium bromide (ET Br). The amplified pattern was visualized on the UV transilluminator and photographed by gel documentation system.

PCR Primers:

For this study, random primers of Operon B series (U.S.A.) were used. Initially 20 primers (decamer) were selected but finally only 06 random primers (Table 1) were used because these primers were found to be effective to produced good quality bands with high resolution.

PCR Amplification

The reaction mixture for RAPD-PCR amplification (25 μ l) was composed of 2.5 μ l of 10X Taq polymerase buffer, 1 μ l of 10mM dNTP, 1.5 μ l of 10 pmole RAPD primer, 0.5 μ l 1U Taq polymerase, 17.5 μ l DD water, 2 μ l of 50mg/ μ l template DNA. A negative control, without template DNA was also included in each round of reactions. After preheating for 5 minute at 94°C, PCR was run for 40 cycles. It consisted of a 94°C denaturation step (1.0 min), 32°C annealing step (1.0 min) and 72°C elongation step (1.5 min) in a gradient thermal cycler (Eppendroff). At the end of the run, a final extension period was appended (72°C, 10 min) and stored at 4°C until the PCR products were analyzed.

Agarose gel Electrophoresis:

The amplified DNA fragments were separated on 1.2% agarose gel (Himedia) and stained with ethidium bromide. The amplified pattern was visualized on the UV trans-illuminator and photographed by gel documentation system.

RAPD Analysis:

RAPD fragments were analyzed from image which was taken gel documentation system. Reproducible bands were scored. The genotype was evaluated on the basis of presence and absence of amplified bands. The percentage of polymorphic and monomorphic loci was calculated. Each

amplified fragments represented a single locus. Genetic Dissimilarity was calculated on the basis of polymorphic loci. Percentage of polymorphism and monomorphism were calculated using the formulas: polymorphic loci (%) = no. of polymorphic loci X 100/total no. of amplified fragments; monomorphic loci (%) = No. of monomorphic loci X 100/total no. of amplified fragments.

3.3. Molecular techniques

The idea of a standardized molecular identification system emerged progressively during the 1990s with the development of PCR-based approaches for species identification. Molecular identification has largely been applied to bacterial studies, microbial biodiversity surveys (e.g. Woese 1996; Zhou et al., 1997) and routine pathogenic strains diagnoses (e.g. Maiden et al., 1998; Wirth et al., 2006) due to a need for culture independent identification systems. PCR-based methods have also been frequently used in fields related to taxonomy, food and forensic molecular identification and for identification of eukaryotic pathogens and vectors (Frezal and Leblois 2008). Both nuclear and mitochondrial genes have been targeted for species identification by PCR techniques.

3.4 Randomly Amplified Polymorphic DNA

(RAPD) RAPD-PCR technique was developed by two independent research groups such as William et al., (1990) and Welsh and Maclelland (1990). The method is simple and quick to perform, and most importantly, prior knowledge of the genetic make-up of the organism in question is not required (Hadrys et al., 1992). The main drawback with RAPD is that homozygous and heterozygous states cannot be differentiated and the patterns are very sensitive to slight change in amplification conditions, giving problems of reproducibility (Ferguson et al., 1995). Other advantage of RAPD is the ease with which a large number of loci and individuals can be screened. RAPD markers have been used in species identification in fishes (Partis and Wells, 1996) and mollusks (Klinbunga et al., 2000; Crossland et al., 1993) analysis of population structure in black tiger shrimp (Tassanakajon et al., 1998) and analysis of genetic diversity (Yue et al., 2002).

Dinesh et al., (1996), in their study used random amplified polymorphic DNA (RAPD) fingerprinting for estimating genetic variation and species differentiation in three species of tilapia. A 16-mer random primer generated RAPD markers ranging from 250 to 2400 base pairs. A total of 13 RAPD markers differentiating the three species of tilapia were detected. This study presented RAPD markers as a new class of useful genetic markers for assessment of genetic diversity and species differentiation in tilapia. Caccone et al. (1997) used RAPD-PCR assays to identify DNA polymorphism in the European sea bass (*Dicentrarchus labrax*). They observed 107 RAPD bands by using eight primers in 260 individuals and found higher variation within population but quite lower in between populations.

Joseph et al., (1996), studied on the genetic variation of four migratory populations of Atlantic Coast striped bass (*Morone saxatilis*) using RAPD technique. Sufficient levels of variation were detected to allow a population genetic analysis of these four populations. In their study significant heterogeneity was detected in the frequencies of 32 percent of surveyed RAPD markers. Results of the RAPD analysis suggest that gene flow is sufficient to prevent fixation of alternate genetic markers, but not sufficient to prevent the development of significant divergence in frequencies of these markers. Jayashankar and Dharmalingam, (1997), extracted genomic DNA from 30 different specimens of Indian mackerel *Rastrelliger kanagurta* (Cuvier) and 5 king seer (Lacepede) by a single modified phenol-chloroform protocol. Employed 10-mer arbitrary primers with the DNA samples produced varying RAPD patterns and those with different sequence generated different banding patterns for the same template in the two commercially important Scombroid fishes. Primers OPA 07 (GAAACGGGTG) yielded more polymorphic loci with high reproducibility.

3.5. Mitochondrial (*COI* and *CYT b*) DNA over Nuclear DNA

The detection of nuclear gene is very useful in species identification but their presence of limited copies within a cell and their inability to recover it from the degraded samples make mitochondrial gene more favorable. There are several advantages of mitochondrial DNA over nuclear DNA. The advantages

are as follows: (a) there is no recombination since most of them are maternal inheritance, with the exception of mutations, and all loci will be linked (b) there is slight DNA that is non-coding, nor are there introns or pseudogenes (c) all the coding sections coding for proteins or RNA molecules are involved in respiration so there is conservation of sequence as any change in the proteins or RNA molecules could adversely affect the organism and (d) No proofreading enzyme exists in the mitochondria to correct DNA bases added incorrectly during DNA replication unlike nuclear DNA. Therefore, the accumulation of single base changes is up to five times higher in mtDNA compared to errors due to DNA replication in nuclear DNA.

3.6. Mitochondrial DNA

Mitochondrial DNA (mtDNA) analysis has been employed in the evolutionary study of animal species for more than 30 years (Brown et al., 1979; Avise and Walker 1999). Each mitochondrion contains several such circular molecules and, therefore, several complete sets of mitochondrial genes. Furthermore, each cell has several mitochondria. When sample tissue is limited, the mitochondrion offers a relatively abundant source of DNA. The molecule is maternally inherited in vertebrates (Giles et al., 1980; Gyllenstein et al., 1985) and reveals little, if any, evidence of recombination. Relative to the nuclear genome, mtDNA evolves at a faster rate (Wilson et al., 1985) with different regions of the genome displaying a wide array of rates, thus making the molecule ideal for within- and between-species comparisons. Recent developments in DNA-sequencing technology have introduced the possibility of using variations in short sequences of DNA as labels for species in a process that has become known as DNA barcoding. The concept has already gained considerable acceptance among those working with species refractory to morphological identification such as bacteria (Urwin and Maiden 2003), protists (Cavalier-Smith 2004) and Rhodophyta (Saunders 2005).

3.7. DNA Barcode

DNA barcoding has been proposed as a useful technique within many disciplines (e.g., conservation biology, medical, forensics and alimentary fields) for determining the taxonomic identity of a sample based on nucleotide

similarity to samples of known taxonomy. DNA barcoding promises fast, accurate species identification by focusing analysis on a short standardized segment of the genome (Hebert et al. 2003). The use of nucleotide sequence variations to investigate evolutionary relationships is not a new concept. Carl Woese used sequence differences in ribosomal RNA (rRNA) to discover archaea, which in turn led to the redrawing of the evolutionary tree.

3.8. DNA barcoding of fishes

Yoo et al. (2006) tested the effectiveness of a COI barcode in discriminating Korean bird species. The 5' terminus of the COI barcode for 92 species of fishes was determined and found unambiguous species identification. The genetic differences between closely related species were on average 25 times higher than the differences within species. Only one misidentified species out of 239 specimens in a genetic resource bank was identified confirming the accuracy of species identification in the banking system. Two potential composite species were also identified. This finding of large COI sequence differences between species confirms the effectiveness of COI barcodes for identifying Korean bird species.

Tavares and Baker (2008) compared ~650 bp of COI in 60 sister-species pairs identified in multigene phylogenies from 10 orders of birds. In all pairs, individuals of each species were monophyletic in a neighbor-joining (NJ) tree, and each species possessed fixed mutational differences distinguishing them from their sister species. Consequently, individuals were correctly assigned to species using a statistical coalescent framework. A coalescent test of taxonomic distinctiveness based on chance occurrence of reciprocal monophyly in two lineages was verified in known sister species, and used to identify recently separated lineages that represent putative species.

3.9 Construction of phylogenetic tree

A phylogeny is a representation of the evolutionary history of a group of related organisms. Now a day molecular data consisting of nucleotides differences among homologous DNA sequences can be used to reconstruct molecular phylogenetics. RAPD permits very rapid generation of large number

of markers needed for gene mapping studies (Liu et al., 1999 and Postlethwait et al., 1994). The RAPD markers follow Mendelian inheritance and are highly reproducible with size ranges of 400-1500bp (Liu et al., 1999 and Postlethwait et al., 1994). These characteristics led many workers to create genetic maps of various fishes even though RAPD markers are dominant alleles. This is a disadvantage of RAPD analysis because markers fail to distinguish between heterozygous and homozygous individuals possessing specific allele.

The potential use of RAPD analysis in phylogenetic studies and population genetics has been documented in a wide variety of organisms (Hadrys et al., 1992). It has been applied to the study of phylogenetic relationships in tilapia and cichlid species (Bardakci and Skibinski, 1994). The technique was found to be useful in reconstructing cichlid fish phylogeny (Sultmann et al., 1995). RAPD has also been used in the identification of largemouth bass subspecies and its intergrades (Williams et al., 1990; Welsh and McCllland, 1990), and for the study of Atlantic coast striped bass (Bielawski et al., 1995). The RAPD technique demonstrated low levels of polymorphism among strains of channel catfish, *Ictalurus punctatus* and Blue catfish *Ictalurus furcatus*, but polymorphism between them was shown to be higher (Liu et al., 1999). Genetic differentiation within and among sea bass *D. labrax* from different locations using RAPD showed high levels of polymorphism (Caccone et al., 1997). Thus RAPD has been used in populations' studies in fisheries, and can be used efficiently for geographic analysis of populations with differential degrees of geographic isolation. In aquatic shellfish, this technique has successfully analyzed genetic polymorphisms at the interspecies level of bivalves, such as the oyster (Klinbunga et al., 2000) and the mussel (Rego et al., 2002) and at the intraspecies level of the Scallop (Patwary et al., 1994), the abalone (Huang et al., 2000) and the oyster (Klinbunga et al., 2001).

Phang et al, (1995) also studied inheritance of RAPD markers in the guppy fish, *Poecilia reticulata*. Three oligonucleotide primers (two 10-mer and one 9-mer) and their paired combinations were found to generate different reproducible RAPD fingerprints in guppies. One of the two 10-mer primers was used to detect DNA polymorphism in 2 guppy varieties. The RAPD markers exhibited dominant Mendelian inheritance and could be used for subsequent

linkage mapping. Sluttmann et al., (1995) studied phylogenetic analysis of cichlid fishes using the random amplification of polymorphic DNA (RAPD) technique to obtain sequence difference between selected cichlid species. They designed specific primers based on these sequences and used them to amplify template DNA from a large number of species by the polymerase chain reaction (PCR).

They sequenced the amplified products and searched the sequences for indels and shared substitutions. They identified a number of such characters at three loci DXTU1, DXTU2 and DXTU3- and used them for phylogenetic and cladistic analysis of the relationships among the various cichlid groups. Their studies assign an outgroup position to Neotropical cichlids in relation to African cichlids, provide evidence for a sister-group relationship of tilapiines to the haplochromines, group *Cyphotilapia frontosa* with the lamprologines of Lake Tanganyika, place *Astatoreochromis alluaudi* to an outgroup position with respect to other haplochromines of Lake Victoria haplochromines and the Lake Malawi haplochromines. The described approach holds great promise for future resolution of cichlid phylogeny.

3.10. Phylogenetic analysis

The study four economic animals (Buffalo, cattle, sheep and goat) belonged to the family Bovidae while pig belonged to the family Suidae. With the advent of molecular techniques, the complicated phylogenetic work of Bovidae has been minimized to some extent. . In recent years, mitochondrial DNA (mtDNA) is widely using for genetic studies to estimate the phylogenetic relationship among species (Hiendleder et al., 1998; Lin et al., 1999; Manceau et al., 1999; San Mauro et al., 2004; Zink 2004; Kan et al., 2010). Many genes in mitochondrial genome have been extensively used for phylogenetic studies (MdZain et al., 2010). They are the cytochrome b gene (Rosli et al., 2011), the 16S rRNA gene (Guo et al., 2012), the 12S rRNA gene (Vun et al. 2011) and the mtDNA control region (Vun et al., 2011)

Saitou and Nei (1987) proposed a new method called the neighbor-joining method for reconstructing phylogenetic trees from evolutionary distance data. The principle of this method is to find pairs of operational taxonomic units

(OTUs [= neighbors]) that minimize the total branch length at each stage of clustering of OTUs starting with a star like tree. The branch lengths as well as the topology of a parsimonious tree can quickly be obtained by using this method. Using computer simulation, the efficiency of this method was study in obtaining the correct unrooted tree in comparison with that of five other tree-making methods: the unweighted pair group method of analysis, Farris's method, Sattath and Tversky's method, Li's method, and Tatenio et al.'s modified Farris method. The new, neighbor-joining method and Sattath and Tversky's method are shown be generally better than the other methods.

Allard et al. (1992) found that nine additional sequences from representatives of different tribes of the family Bovidae were combined with six published artiodactyls sequences to provide orthologous mtDNA for investigation of bovid phylogeny and evolution. Each species was represented by a homologous 2.7-kilobase-pair stretch of mtDNA for the complete 12S and 16S rRNA genes and three adjacent tRNA genes. These data, when compared to their results, provided evidence for a monophyletic Bovidae and for two clades within the family: one includes the tribes Boselaphini, Bovini, and Tragelaphini and another for an Antilopini/ Neotragini grouping. All other intrafamilial relationships were only weakly supported. These sequence comparisons suggest that most bovid tribes originated early in the Miocene with all extant lineages present by 16-17 million years ago. Thus, bovid tribes provide an example of rapid cladogenesis, following the origin of families in the infra-order Pecora. Grossi et al. (2006) examined the variation in mitochondrial DNA by sequencing the D-loop region in wild and domestic (large-white breed) pigs, in hybrids between domestic and wild pigs, and in Monteiro pigs. This study showed European Haplotypes in Monteiro pigs and wild pigs while domestic pigs had both European and Asian haplotypes

Bunch et al. (2006) analyzed mitochondrial cytochrome b gene sequence to determine the history of Ovis. Based on this gene, the history of true sheep (Ovis) began approximately 3.12 million years ago (MYA). The evolution of Ovis resulted in three generally accepted genetic groups: Argaliforms, Moufloniforms, and Pachyceriforms. The Pachyceriforms of the subgenus Pachyceros comprise the thin horn sheep, Ovis nivicola (snow sheep), Ovis dalli

(Dall and Stone sheep), and *Ovis canadensis* (Rocky Mountain and desert bighorn). North America wildsheep (*O. canadensis* and *O. dalli*) evolved separately from Eurasian wild sheep and diverged from each other about 1.41 MYA. Ancestral stock that gave rise to snow sheep, Moufloniforms, and Argaliforms occurred 2.3 MYA, which then gave rise to two different extant lines of snow sheep that diverged from each other about 1.96 MYA. The more recent nivicola line is genetically closer to the North American wild sheep and may represent a close association during the refugium when Alaska and Siberia were connected by the Bering land bridge. The earlier period of evolution of the Pachyceriforms suggests they may have first evolved in Eurasia, the oldest ancestor then giving rise to North American wild sheep, and that a canadensislike ancestor most likely gave rise to nivicola. Meadows et al. (2007) sequenced a total of 2027 bp of mitochondrial DNA (mtDNA) from 197 sheep revealed a total of 85 haplotypes and a high level of genetic diversity. Six individuals carried three haplotypes, which clustered separately from the known ovine mtDNA lineages A, B, and C. Analysis of genetic distance, mismatch distribution, and comparisons with wild sheep confirmed that these represent two additional mtDNA lineages denoted D and E. The two haplogroup E sequences were found to link the previously identified groups A and C. The single haplogroup D sequence branched with the eastern mouflon (*Ovis orientalis*), urial (*O. vignei*), and argali (*O. Ammon*) sheep. High sequence diversity ($K = 1.86\%$, haplogroup D and *O.orientalis*) indicates that the wild progenitor of this domestic lineage remains unresolved. The identification in this study of evidence for additional domestication events adds to the emerging view that sheep were recruited from wild populations multiple times in the same way as for other livestock species such as goat, cattle, and pig. Gupta et al. (2013) described genetic differentiation of Indian wild and domestic pigs using a molecular technique. The generated partial fragment (421bp) of mitochondrial DNA (mtDNA) cytochrome b (Cyt b) gene analysis exhibited unambiguous (>3%) genetic variation between Indian wild and domestic pigs. Nine forensically informative nucleotide sequence (FINS) variations between Indian wild and domestic pigs were observed. This study indicates that domestic pigs in India are not descendent of the Indian wild pig, however; they are closer to

the other wild pig races found in Asia and Europe. Nijman and Aliabadian (2013) analyzed data from the *cox1* gene for 31 individuals of seven subspecies of the barn owl *Tyto Alba*, and compared this with 214 sequences from 29 other owl species. Phylogenetic analysis of the *T. Alba* samples gives very strong support for an Old World alba-clade (three subspecies) and a New World furcata-clade (four subspecies) that are genetically equidistant. The amount of intraspecific variation within each of these clades ranges from 0.66–0.99%, but variation among these clades ranges from 5.33–6.20%. These combined data suggested that barn owl of the Old World is indeed best considered a separate species different from that of the New World.

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IV.RESULT AND DISCUSSION

The fish fauna and their ecological status were surveyed from the streams and rivers of the Southern Western Ghats during march-2020 to Dec-2020. The studying sites were selected based on the earlier faunal distribution in literature. Ghats is a mountain range that runs almost parallel to the Western Ghats of the Indian peninsula well known for its splendid biodiversity both in terms of richness and endemism. The physical and climatic records of the selected study sites (Adukkam, Bear Shola Falls, Fairy falls, Guntar, Kozikottu, Kumbakarai, Kurusedi, Moolayar, Pachalur, Pillar Rock, Poolathur, Rat tail falls, Samakadu, Silver cascade, and thalakuthu falls is given in chemical parameters recorded from the study sites maximum water temperature 24°C was recorded at Poolathur and a minimum water temperature (13.4°C) was noted at Guntar. Characteristic features of small streams such as stream width, water depth, current velocity, and percentage canopy cover, bedrock, boulder, pebbles, and sand in the study sites are affected more human disturbance before pandemic situation. In large-scale studies performed in this areas during pandemic situation , Geo-morphological variables such as Acid base(PH), Dissolved oxygen level(DO₂),Total dissolved salt(TDS) and Salinity not affected by human activity and climate change have been considered as the major factors responsible for fish faunal,vegetational distribution was increased. Results indicate that variables for determining the diversity of fish communities. Multivariate analysis suggests that physical and chemical parameters variables significantly influence the distribution of fish community and abundance fish fauna in the streams of palani hills.

**TABLE 1: COLLECTION STREAMS WITH ELEVATION IN
PALANI HILLS OF SOUTH INDIA.**

S.NO	Low elevation(m)	S.NO	Mid elevation(m)	S.NO	High elevation (m)
1.	Poolathur (900)	6	Adukkam (1406)	11.	Fairy falls (2050)
2.	Kumbakarai (400)	7	Silver cascade (1256)	12.	Guntar (1990)
3.	Pachalur (296)	8.	Kurusedi (1213)	13.	Bear Shola falls (1980)
4.	Rat tail falls (290)	9.	Moolayar (1186)	14.	Samakadu (1538)
5.	Thallakuttu (250)	10.	Kozikottu (1182)	15.	Pillar rock stream (1534)

**Table 1 PHYSICAL PARAMETERS OF THE SAMPLING
SITES: PRE MONSOON (2019)**

Area of collection	Latitude (n)	Longitude (e)	Elevation (m)	Temperature		Stream width (m)	Water depth (cm)	Current velocity	Canopy cover	Local human distribution
				At	Water					
1.Poolathur	10.16152	77.321	900	17.2	13.4	2	11	0.09	90	YES
2.Kumbakarai	10.17739	77.535	400	17.2	14.8	8	7	10.7	80	YES
3.Pachalur	10.39085	77.667	296	17.3	15.2	4	5	2.1	80	YES
4.Rat tail falls	10.13133	77.373	290	22.7	16.5	9	5	0.08	60	YES
5.Thallakuttu falls	10.44181	77.686	250	19.7	16.7	5.5	40	0.09	50	YES
1.Adukkam	10.23677	77.543	1406	23.5	18.4	1406	3.5	3.9	10	NO
2.Silver cascade	10.24537	77.516	1256	22.1	19.7	3	10	0.03	80	NO
3.Kurusedi	10.27633	77.559	1213	22.3	20.1	2	5	50.9	80	YES
4.Moolayar	10.26903	77.611	1186	22.4	20.1	5	6	10.05	50	YES
5.Kozikottu	10.23731	77.546	1182	27.3	20.6	2	15	3.2	50	YES
1.Fairy falls	10.13524	77.281	2050	23.4	21.5	2	15	0.09	40	YES
2.Guntar	10.10927	77.318	1990	28.0	22.5	5	11	0.1	60	YES
3.Bear shoal falls	10.13182	77.239	1980	24.6	23.1	2	20	0.04	60	YES
4.Samakadu	10.24044	77.564	1538	26.7	23.2	2	3	3.2	20	YES
5.Pillar rock	10.21018	77.466	1534	26.0	24.1	2	5	2.1	50	NO

CHEMICAL PARAMETERS OF THE SAMPLING SITES

Area of collection	SALINITY (PPM)	TDS (Total Dissolved Solids)ppm	PH (Acid base valve)	EC (Electrical Conductivity)	DO ₂ (mgL ⁻¹)
1.Poolathur	42.2	48.5	7.1	55.3	8.2
2.Kumbakarai	43.1	90.3	7.6	75.1	8.1
3.Pachalur	63.2	95.2	6.8	122.3	10.2
4.Rat tail falls	48.8	67.2	7.4	82.4	7.5
5.Thallakuttu falls	75.2	126.4	6.7	432.2	9.3
1.Adukkam	13.6	20.6	6.7	129.3	13.5
2.Silver cascade	12.1	18.4	6.3	26.4	6.3
3.Kurusedi	28.7	17.5	6.3	44.7	5.6
4.Moolayar	91.6	67.2	7.2	63.9	7.4
5.Kozikottu	36.2	44.2	7.3	62.4	13.6
1.Fairy falls	56.2	78.1	7.6	100.2	8.3
2.Guntar	63.4	55.5	6.9	75.6	13.2
3.Bear shoal falls	13.7	50.7	7.3	98.3	8.9
4.Samakadu	15.7	13.6	7.1	18.9	7.3
5.Pillar rock	17.5	21.1	6.2	30.5	6.9

Table 2. PHYSICAL PARAMETERS OF THE SAMPLING SITES: MONSOON SEASON (2019)

Area of collection	Latitude (n)	Longitude (e)	Elevation (m)	Temperature		Stream width (m)	Water depth (cm)	Current velocity	Canopy cover	Local human distribution
				At	Water					
1.Poolathur	10.16152	77.321	900	17.2	13.4	2	11	0.09	90	YES
2.Kumbakarai	10.17739	77.535	400	17.2	14.8	8	7	10.7	80	YES
3.Pachalur	10.39085	77.667	296	17.3	15.2	4	5	2.1	80	YES
4.Rat tail falls	10.13133	77.373	290	22.7	16.5	9	5	0.08	60	YES
5.Thallakuttu falls	10.44181	77.686	250	19.7	16.7	5.5	40	0.09	50	YES
1.Adukkam	10.23677	77.543	1406	23.5	18.4	1406	3.5	3.9	10	NO
2.Silver cascade	10.24537	77.516	1256	22.1	19.7	3	10	0.03	80	NO
3.Kurusedi	10.27633	77.559	1213	22.3	20.1	2	5	50.9	80	YES
4.Moolayar	10.26903	77.611	1186	22.4	20.1	5	6	10.05	50	YES
5.Kozikottu	10.23731	77.546	1182	27.3	20.6	2	15	3.2	50	YES
1.Fairy falls	10.13524	77.281	2050	23.4	21.5	2	15	0.09	40	YES
2.Guntar	10.10927	77.318	1990	28.0	22.5	5	11	0.1	60	YES
3.Bear shoal falls	10.13182	77.239	1980	24.6	23.1	2	20	0.04	60	YES
4.Samakadu	10.24044	77.564	1538	26.7	23.2	2	3	3.2	20	YES
5.Pillar rock	10.21018	77.466	1534	26.0	24.1	2	5	2.1	50	NO

CHEMICAL PARAMETERS OF THE SAMPLING SITES

Area of collection	SALINITY (PPM)	TDS (Total Dissolved Solids)ppm	PH (Acid base valve)	EC (Electrical Conductivity)	DO ₂ (mgL ⁻¹)
1.Poolathur	42.2	48.5	7.1	55.3	8.2
2.Kumbakarai	43.1	90.3	7.6	75.1	8.1
3.Pachalur	63.2	95.2	6.8	122.3	10.2
4.Rat tail falls	48.8	67.2	7.4	82.4	7.5
5.Thallakuttu falls	75.2	126.4	6.7	432.2	9.3
1.Adukkam	13.6	20.6	6.7	129.3	13.5
2.Silver cascade	12.1	18.4	6.3	26.4	6.3
3.Kurusedi	28.7	17.5	6.3	44.7	5.6
4.Moolayar	91.6	67.2	7.2	63.9	7.4
5.Kozikottu	36.2	44.2	7.3	62.4	13.6
1.Fairy falls	56.2	78.1	7.6	100.2	8.3
2.Guntar	63.4	55.5	6.9	75.6	13.2
3.Bear shoal falls	13.7	50.7	7.3	98.3	8.9
4.Samakadu	15.7	13.6	7.1	18.9	7.3
5.Pillar rock	17.5	21.1	6.2	30.5	6.9

Table 3. PHYSICAL PARAMETERS OF THE SAMPLING SITES: WINTER SEASON (2019)

Area of collection	Latitude (n)	Longitude (e)	Elevation (m)	Temperature		Stream width (m)	Water depth (cm)	Current velocity	Canopy cover	Local human distribution
				At	Water					
1.Poolathur	10.16152	77.321	900	17.2	13.4	2	11	0.09	90	YES
2.Kumbakarai	10.17739	77.535	400	17.2	14.8	8	7	10.7	80	YES
3.Pachalur	10.39085	77.667	296	17.3	15.2	4	5	2.1	80	YES
4.Rat tail falls	10.13133	77.373	290	22.7	16.5	9	5	0.08	60	YES
5.Thallakuttu falls	10.44181	77.686	250	19.7	16.7	5.5	40	0.09	50	YES
1.Adukkam	10.23677	77.543	1406	23.5	18.4	1406	3.5	3.9	10	NO
2.Silver cascade	10.24537	77.516	1256	22.1	19.7	3	10	0.03	80	NO
3.Kurusedi	10.27633	77.559	1213	22.3	20.1	2	5	50.9	80	YES
4.Moolayar	10.26903	77.611	1186	22.4	20.1	5	6	10.05	50	YES
5.Kozikottu	10.23731	77.546	1182	27.3	20.6	2	15	3.2	50	YES
1.Fairy falls	10.13524	77.281	2050	23.4	21.5	2	15	0.09	40	YES
2.Guntar	10.10927	77.318	1990	28.0	22.5	5	11	0.1	60	YES
3.Bear shoal falls	10.13182	77.239	1980	24.6	23.1	2	20	0.04	60	YES
4.Samakadu	10.24044	77.564	1538	26.7	23.2	2	3	3.2	20	YES
5.Pillar rock	10.21018	77.466	1534	26.0	24.1	2	5	2.1	50	NO

CHEMICAL PARAMETERS OF THE SAMPLING SITES

Area of collection	SALINITY (PPM)	TDS (Total Dissolved Solids)ppm	PH (Acid base valve)	EC (Electrical Conductivity)	DO ₂ (mgL ⁻¹)
1.Poolathur	42.2	48.5	7.1	55.3	8.2
2.Kumbakarai	43.1	90.3	7.6	75.1	8.1
3.Pachalur	63.2	95.2	6.8	122.3	10.2
4.Rat tail falls	48.8	67.2	7.4	82.4	7.5
5.Thallakuttu falls	75.2	126.4	6.7	432.2	9.3
1.Adukkam	13.6	20.6	6.7	129.3	13.5
2.Silver cascade	12.1	18.4	6.3	26.4	6.3
3.Kurusedi	28.7	17.5	6.3	44.7	5.6
4.Moolayar	91.6	67.2	7.2	63.9	7.4
5.Kozikottu	36.2	44.2	7.3	62.4	13.6
1.Fairy falls	56.2	78.1	7.6	100.2	8.3
2.Guntar	63.4	55.5	6.9	75.6	13.2
3.Bear shoal falls	13.7	50.7	7.3	98.3	8.9
4.Samakadu	15.7	13.6	7.1	18.9	7.3
5.Pillar rock	17.5	21.1	6.2	30.5	6.9

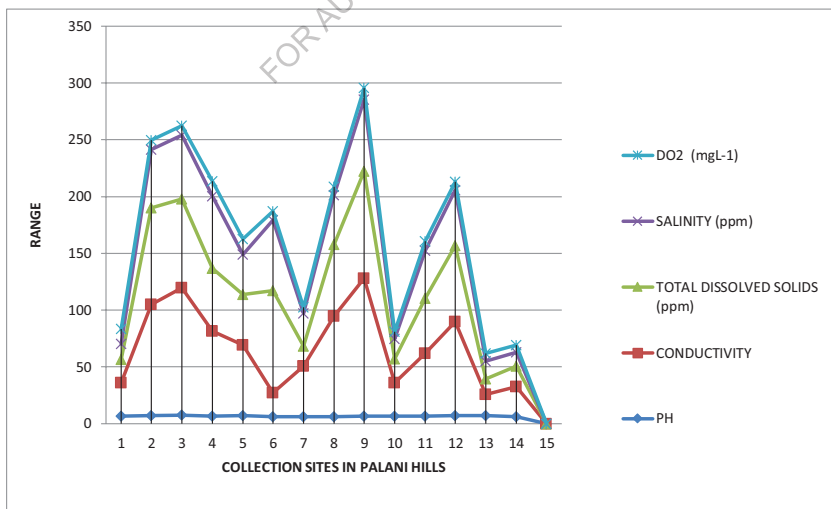
TABLE 4. PHYSICAL PARAMETERS OF THE SAMPLING SITES: SUMMER SEASON (2019)

Area of collection	Latitude (n)	Longitude (e)	Elevation (m)	Temperature		Stream width (m)	Water depth (cm)	Current velocity	Canopy cover	Local human distribution
				At	Water					
1.Poolathur	10.16152	77.321	900	17.2	13.4	2	11	0.09	90	YES
2.Kumbakarai	10.17739	77.535	400	17.2	14.8	8	7	10.7	80	YES
3.Pachalur	10.39085	77.667	296	17.3	15.2	4	5	2.1	80	YES
4.Rat tail falls	10.13133	77.373	290	22.7	16.5	9	5	0.08	60	YES
5.Thallakuttu falls	10.44181	77.686	250	19.7	16.7	5.5	40	0.09	50	YES
1.Adukkam	10.23677	77.543	1406	23.5	18.4	1406	3.5	3.9	10	NO
2.Silver cascade	10.24537	77.516	1256	22.1	19.7	3	10	0.03	80	NO
3.Kurusedi	10.27633	77.559	1213	22.3	20.1	2	5	50.9	80	YES
4.Moolayar	10.26903	77.611	1186	22.4	20.1	5	6	10.05	50	YES
5.Kozikottu	10.23731	77.546	1182	27.3	20.6	2	15	3.2	50	YES
1.Fairy falls	10.13524	77.281	2050	23.4	21.5	2	15	0.09	40	YES
2.Guntar	10.10927	77.318	1990	28.0	22.5	5	11	0.1	60	YES
3.Bear shoal falls	10.13182	77.239	1980	24.6	23.1	2	20	0.04	60	YES
4.Samakadu	10.24044	77.564	1538	26.7	23.2	2	3	3.2	20	YES
5.Pillar rock	10.21018	77.466	1534	26.0	24.1	2	5	2.1	50	NO

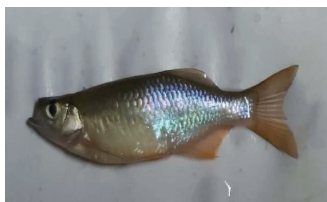
CHEMICAL PARAMETERS OF THE SAMPLING SITES

Area of collection	SALINITY (PPM)	TDS (Total Dissolved Solids)ppm	PH (Acid base valve)	EC (Electrical Conductivity)	DO ₂ (mgL ⁻¹)
1.Poolathur	42.2	48.5	7.1	55.3	8.2
2.Kumbakarai	43.1	90.3	7.6	75.1	8.1
3.Pachalur	63.2	95.2	6.8	122.3	10.2
4.Rat tail falls	48.8	67.2	7.4	82.4	7.5
5.Thallakuttu falls	75.2	126.4	6.7	432.2	9.3
1.Adukkam	13.6	20.6	6.7	129.3	13.5
2.Silver cascade	12.1	18.4	6.3	26.4	6.3
3.Kurusedi	28.7	17.5	6.3	44.7	5.6
4.Moolayar	91.6	67.2	7.2	63.9	7.4
5.Kozikottu	36.2	44.2	7.3	62.4	13.6
1.Fairy falls	56.2	78.1	7.6	100.2	8.3
2.Guntar	63.4	55.5	6.9	75.6	13.2
3.Bear shoal falls	13.7	50.7	7.3	98.3	8.9
4.Samakadu	15.7	13.6	7.1	18.9	7.3
5.Pillar rock	17.5	21.1	6.2	30.5	6.9

GRAPH: 1. STATISTICAL ANALYSIS PHYSICO CHEMICAL PARAMETERS OF COLLECTION SITES:



**COLLECTION OF FISHES IN PALANI HILLS (WESTERN
GHATS) STREAMS**



V.CONCLUSION:

In large-scale studies performed in this areas during pandemic situation , Geo-morphological variables such as Acid base(PH), Dissolved oxygen level(DO₂),Total dissolved salt(TDS) and Salinity not affected by human activity and climate change have been considered as the major factors responsible for fish faunal,vegetational distribution was increased. Results indicate that variables for determining the diversity of fish communities. Multivariate analysis suggests that physical and chemical parameters variables significantly influence the distribution of fish community and abundance fish fauna in the streams of palani hills. The proposed study would be helpful to understand the status of the distributions climatic changes in Palani hills streams. This study will be useful to carry physic-chemical parameters of streams of Palani hills of South India. This study, the water quality assessment of freshwater of streams and rivers will be made living of endemic freshwater fishes and its bio-ecology is scarce and received little attention during Covid-19 pandemic situation, the analysis study would be useful to understand the conservation status of freshwater streams of Palani hills and its endemic freshwater fishes.

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